

# City Energy Analyst (CEA): Integrated framework for analysis and optimization of building energy systems in neighborhoods and city districts



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## ABSTRACT

This paper describes the City Energy Analyst (CEA), a computational framework for the analysis and optimization of energy systems in neighborhoods and city districts. The framework allows analyzing the energy, carbon and financial benefits of multiple urban design scenarios in conjunction to optimal schemes of distributed generation. For this, the framework integrates time-dependent methods for building energy performance simulation, conversion and storage technologies simulation, assessment of local energy potentials, bi-level energy systems optimization and multi-criteria analysis. Based on past research, the framework introduces a novel interface to facilitate the spatiotemporal analysis of patterns of demand and potential infrastructure solutions. The model was programmed in Python v2.7 and built as an extension of the Geographic Information System ArcGIS v10.3, which serves as a platform for the allocation and future dissemination of spatiotemporal data. We present an application of the model for a downtown area in Switzerland where we evaluated four trajectories of development and found optimum infrastructure solutions for their operation. For a more holistic approach we used the 2000-W/1-t CO<sub>2</sub> society vision concept to compare the environmental performance of these solutions with that of embodied energy in buildings and transportation systems. From the optimization process, most infrastructure solutions showed an average integration of 50% to 80% of buildings in thermal micro-grids, 50 to 100% of the available solar potential, and a resource mix consisting of photovoltaic electricity and sources of waste and ambient heat. For a balanced distribution of social, environmental and economic criteria, the results showed potential relative savings in the area from 45% to 60% in emissions and from 25% to 50% in primary energy at an annualized cost between 14% and 44% higher than today. For an economic-driven distribution, the results showed savings of up to 23% in emissions, 36% in primary energy and 11% in costs. We identified close to 15% in emissions and 20% in primary energy savings with variable costs between –2% and 23% in the area are strongly related to the urban design option rather than to its optimal energy system. In comparison to local benchmarks, the environmental impact of buildings during operation lies between that of embodied energy in buildings and mobility in the service sector (business flights). We estimated that an increase in close to 4% of today's average efficiency of photovoltaic technology would allow the area to comply with those local benchmarks. On the other hand, we concluded the suitability of the City Energy Analyst (CEA) to assist urban planning authorities looking for both design and engineering options to increase the performance of their neighborhoods and city districts.

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## 1. Introduction

In line with contemporary targets of sustainable development, urban planning authorities require a comprehensive view of cost-effective opportunities to improve the performance of buildings at the neighborhood and district scale [1–3]. These opportunities could range from building retrofits (envelope and systems) and distributed generation schemes (thermal and/or electrical

**Abbreviations:** PV, photovoltaic; PV/T, photovoltaic thermal; SC, solar collector; O&M, operation and maintenance; INV, Investment costs; TAC, Total annual costs; CO<sub>2</sub>, Green house gas emissions; RE, reliability; %RES, Percentage of renewables; PEN, primary energy–non-renewable; LEC, levelized energy cost; rME, relative mean error; rRMSE, relative root mean square error; RMSE, root mean square error.

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**Nomenclature***Symbol [unit]*

|              |                                                    |
|--------------|----------------------------------------------------|
| $I$          | isolation [W/m <sup>2</sup> ]                      |
| $\Delta T$   | delta of temperature [°K]                          |
| $Z$          | depth [m]                                          |
| $T$          | temperature [°K]                                   |
| $A$          | area [m]                                           |
| $S$          | spacing [m]                                        |
| $t$          | time [h]                                           |
| $\rho$       | density [kg/m <sup>3</sup> ]                       |
| $C_p$        | specific heat capacity [kg/kJ °K]                  |
| $\dot{V}$    | volumetric flow rate [m <sup>3</sup> /s]           |
| $h$          | height [m]                                         |
| $K$          | thermal conductivity [W/m °K]                      |
| $g$          | gravity [m/s <sup>2</sup> ]                        |
| $\alpha T$   | thermal expansion coefficient [1/°K <sup>2</sup> ] |
| $D$          | diameter [m]                                       |
| $CD$         | drag coefficient [–]                               |
| $\dot{v}$    | velocity [m/s]                                     |
| $\phi$       | energy flow [Wh];                                  |
| $\mu$        | utilization factor [–]                             |
| $C$          | coefficient [–]                                    |
| $\varphi$    | factor [–]                                         |
| $n$          | efficiency [–]                                     |
| $V$          | volume [m <sup>3</sup> ]                           |
| $P$          | pressure [kPa]                                     |
| $\dot{m}$    | mass flow rate [kg/s]                              |
| $L$          | length [m]                                         |
| $R$          | thermal resistance [m <sup>2</sup> °K/W]           |
| wet.p        | wet perimeter [m]                                  |
| $b$          | thickness [m]                                      |
| COP          | coefficient of performance                         |
| $B$          | slope [°]                                          |
| $\theta$     | orientation [°]                                    |
| $\tau\alpha$ | Incident angle [°rad]                              |
| Nu           | Nusselt number [–]                                 |
| Pr           | Prandtl number [–]                                 |
| NTU          | number of transfer units [–]                       |
| $r$          | rugosity [–]                                       |
| $M$          | air-mass modifier [–]                              |
| $W$          | width [m]                                          |
| Rfl          | reflectance                                        |
| Lt           | lifetime [–]                                       |
| ind          | Individual points [–]                              |
| $N$          | number [–]                                         |
| $Re$         | Reynolds number [–]                                |
| $fr$         | Darcy friction coefficient [–]                     |

*Superscript*

|       |         |
|-------|---------|
| 0     | nominal |
| max   | maximum |
| min   | minimum |
| best  | best    |
| worst | worst   |

*Subscript*

|      |             |
|------|-------------|
| hor  | horizontal  |
| $z$  | depth       |
| BH   | borehole    |
| $g$  | ground/soil |
| $a$  | air         |
| $w$  | water       |
| lake | lake        |

|        |                           |
|--------|---------------------------|
| surf   | surface                   |
| bottom | bottom                    |
| 1      | first degree coefficient  |
| 2      | second degree coefficient |
| 3      | third degree coefficient  |
| 4      | fourth degree coefficient |
| wind   | wind                      |
| data   | server-room               |
| th     | thermal energy            |
| el     | electrical energy         |
| pl     | part load                 |
| fan    | fan                       |
| LP     | low pressure              |
| HP     | high pressure             |
| ca     | compressed air            |
| sup    | supply                    |
| re     | return                    |
| exp    | expansion                 |
| comp   | compression               |
| pol    | polytrophic               |
| spur   | spur                      |
| ww     | domestic hot water        |
| fw     | fresh water               |
| $p$    | pipng                     |
| wet    | Wet cooling tower         |
| 0      | $t - 1$ period            |
| sto    | storage                   |
| car    | characteristic            |
| ext    | external                  |
| int    | internal                  |
| ch     | chilled                   |
| sw     | sewage                    |
| gr     | group                     |
| ab     | absorbed                  |
| cell   | cell                      |
| $B$    | beam                      |
| mesh   | mesh                      |
| $b$    | building                  |
| in     | internal                  |
| $H$    | heating vector            |
| $c$    | cooling vector            |
| $E$    | electricity vector        |
| ap     | aperture                  |
| lyr    | layer                     |
| pk     | peak                      |
| dev    | device                    |
| pro    | processes                 |
| $u$    | utility                   |
| vcc    | vapor compression chiller |
| $i$    | $i$ -esimo attribute      |
| $j$    | $j$ -esimo attribute      |
| $k$    | $k$ -esimo attribute      |
| w/gl   | Water–glycol              |
| $D$    | direct                    |
| yr     | year                      |
| an     | annualization             |
| test   | test                      |
| boiler | boiler                    |
| cc     | combined cycle            |
| st     | steam generation          |
| cond   | condensation              |
| DH     | district heating          |
| turb   | turbine                   |
| exh    | exchanger                 |

|        |                      |
|--------|----------------------|
| FC     | fuel cell            |
| m      | module               |
| all    | all                  |
| source | source               |
| sink   | sink                 |
| A      | approach             |
| HP     | Heat pump            |
| aux    | auxiliary            |
| net    | network              |
| region | regional             |
| pv/t   | photovoltaic thermal |
| l      | loss                 |
| sc     | solar collector      |
| pv     | photovoltaic         |
| rq     | required             |

micro-grids) to on-site exploitation of endogenous resources (renewables and waste heat) [4].

In the context of urban transformation, energy planners need to reveal and effectively communicate optimal configurations of those opportunities [5]. These configurations should not only comply with performance targets at financial, energy, carbon and technical levels, but also be visualized to facilitate its discussion and approval [1,6].

In technical terms, the study of optimal energy systems configurations is today a multi-scale, multi-objective and multi-carrier problem. At the neighborhood scale, these problems require a detailed description of spatiotemporal patterns of building demand [1] and resource availability [7], as well as advanced models for simulation of technology performance [8]. Extensive research has been done in approaches for building demand prognosis [9,10] (i.e. top-down [11–15], bottom-up [16–20], and hybrid [1,4,21–23]), resource potential analysis (i.e. for biomass [24–28], solar energy [29–31], wind energy [32–35], ambient heat [36–40], hydropower [41], tidal energy [42–44], and waste heat [25,45,46]), multi-objective, multi-carrier, multi-scale optimization of building energy systems (i.e. [7,47–58]), and visualization of energy qualities (i.e. energy potentials [6,59], and energy demand [1,12,60,61]).

Inline with governmental interests to integrate energy planning and urban design strategies at unison (i.e. [62–64]) an integrated approach for modeling such phenomena becomes necessary. This type of approach should attain reliable predictions and effectively communicate patterns of energy consumption, avail-

ability of resources and optimal energy systems configurations to local stakeholders and decision makers.

Based on past research [1], this paper presents the City Energy Analyst (CEA), an integrated framework to optimize building energy systems at neighborhood and district scales. In contrast to existing approaches, the framework develops in a single interface a series of detailed models for building demand forecasting, resource availability assessment, simulation of conversion, storage and distribution technologies, bi-level optimization, multi-criteria assessment and four-dimensional (4D) visualization.

The framework allows to holistically evaluate options of building and infrastructure retrofit, and find optimal energy generation schemes (both totally or partially distributed, or combinations). In addition, it allows to analyze the energy, carbon and financial benefits of multiple urban scenarios and infrastructure options. The results can be represented in an intuitive interface for both urban and energy planning authorities.

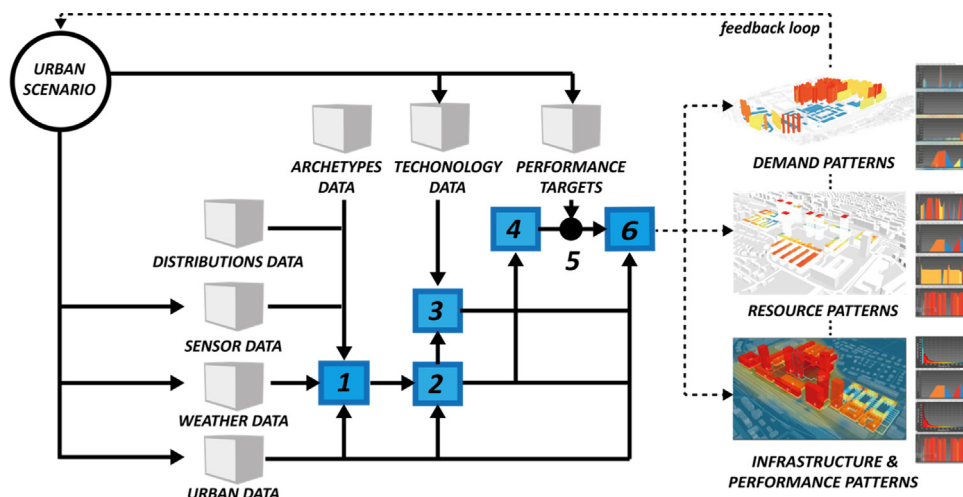
Section 2.2 presents a series of simulation models for assessment of local energy resources. Section 2.3 describes techno-economic models of technologies for conversion, storage, and distribution of thermal energy. Section 2.4 presents a bi-level optimization method for sizing and operating energy systems. Section 2.5 presents a series of methods for the analysis and visualization of energy system's performance. In Section 3 the framework is tested in a real case of urban transformation, for which four urban design scenarios were created. Finally, the paper is concluded in Section 4.

## 2. Methodology

The general framework of the City Energy Analyst (CEA) uses seven databases and six calculation modules (Fig. 1). It is developed in the open-source programming language Python v2.7 and built as an extension of the Geographic Information System ArcGIS v10.3 [65]. Native algorithms of this platform are used in addition to store, analyze and visualize information at spatial and temporal scales.

The weather database stores time-series data of ambient temperature, relative humidity, and solar transmissivity obtained with the software Meteonorm 7.0 [66].

The urban database contains geo-referenced information of buildings, infrastructure, local resources and topography. The data are gathered from official databases and open street maps [67]. It includes building properties such as footprint area, height, window to wall ratios, shading systems types, occupancy types, generation systems and year of construction and last renovation. It also



**Fig. 1.** CEA framework, (1) demand module, (2) resource potential module, (3) systems technology module, (4) system optimization module, (5) decision module, (6) spatiotemporal analysis module.

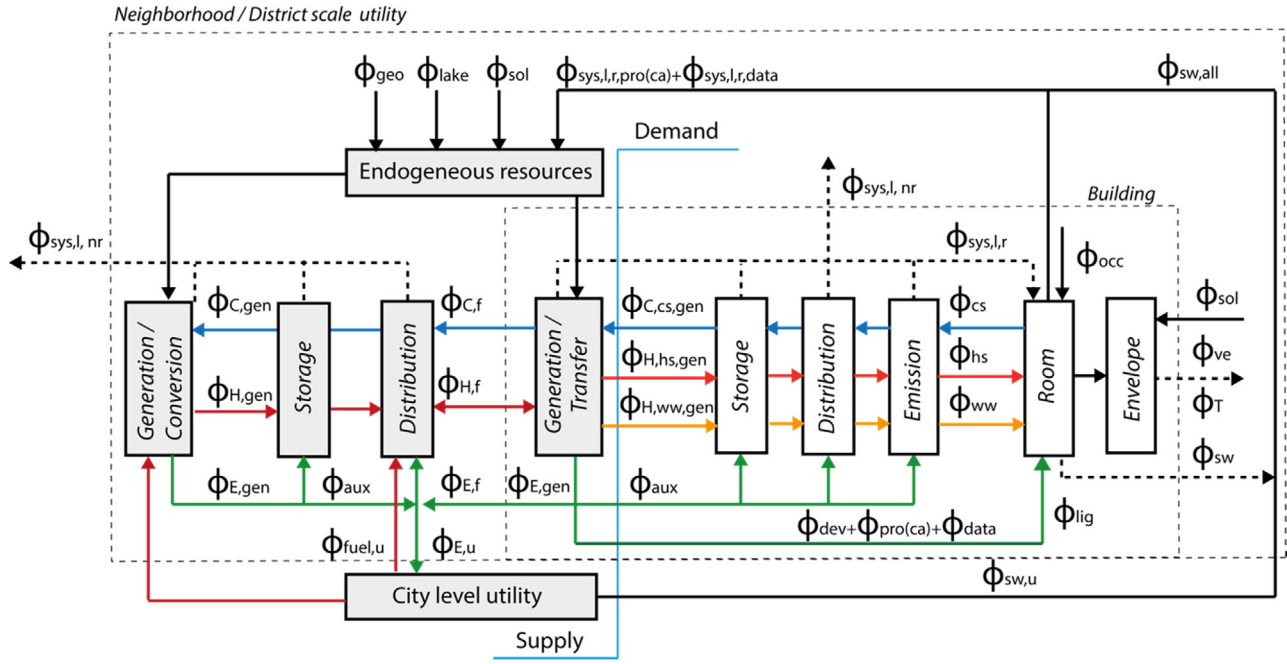


Fig. 2. Simplified energy supply chain. In gray: sub-systems developed in this article. In white: sub-systems covered in Ref. [1].

includes information about location of energy networks, sewage lines, roads, and characteristics of soil and water bodies' such as type and stratification. A digital elevation model describes the topography of the area.

The archetypes database describes standard properties of the local building stock regarding their building envelope and HVAC systems. The database is based on statistical data gathered for buildings constructed before 2005 [68–76]. Values for further periods are obtained from linear interpolation and local building regulations [1].

The distributions database contains standard time-series data for all building types described in the archetypes database. These time-series contain schedules of occupancy, minimum ventilation rates, and temperature and humidity set points for buildings. In addition, it includes standard specific hourly consumption values for appliances, lighting, server-rooms and cold-rooms. It also contains volumetric flow rates requirements of fresh and hot water services. These data is used together with the archetypes database in the demand module of Section 2.1.

The sensor database is a complementary dataset containing data about non-standardized energy services in building uses such as industry and other quantities related to infrastructural components (i.e. mass flow rates, operating temperatures etc.)

The metrics database contains technical and economic metrics of conversion, distribution and storage technologies at the district scale. Abstracts of this database are presented in Table 2, Table 3 and throughout the systems technology module of Section 2.3.

The targets database stores percentages of decision criteria and key performance values for selection of infrastructure options (see Sections 2.5.2 and 2.5.4).

Based on past research [1] we developed a series of calculation modules for simulation of resource availability potential, optimization and visualization of building energy systems from the building to the neighborhood/district scale (Sections 2.1 to 2.6). For this we took into consideration the simplified supply chain of Fig. 2. In this figure, all energy fluxes in the sub-systems for generation, conversion, distribution, emission, and transfer of energy at the district

level are portrayed. Those part of the demand side are described in detail in [1].

## 2.1. Demand module

We calculated the demand of energy in buildings with the dynamic building energy model of [1]. In the model, the final energy demand of buildings is decomposed in manifold end-uses and estimated in an hourly basis at both qualitative (i.e. temperature, voltage requirements) and quantitative levels (i.e. power intensity and mass flow rates). This allows addressing end-uses such as space heating and cooling, domestic hot water, appliances and lighting, as well as more specific end-uses such as server-room cooling, industrial equipment electricity and process heat. Some of these end-uses are part of the input for the model of potential waste heat presented in Section 2.2.3.

## 2.2. Resource potential module

The resource potential module aggregates physical models for the assessment of endogenous energy sources. The estimated accuracy of each model as well as its typical and acceptable values are presented in Table 1.

### 2.2.1. Roof-top solar potential

The hourly solar insolation in rooftop areas is estimated with the hemispherical view-shed algorithm of Fu et al. [35,36], implemented in ArcGIS 10.3. The algorithm models changes in solar trajectory, topographic obstructions (i.e. buildings and terrain), atmospheric effects, latitude and elevation. In order to increase the accuracy of the model, we calibrated parameters such as extraterrestrial transmissivity, level of cloudiness and daylight saving periods to local weather data.

For the analysis, we discretized roof-top areas in  $k$  rectangular surfaces of  $2\text{ m} \times 2\text{ m}$  and calculated the solar insolation at their centroid in the horizontal plane ( $I_{hor,k}$ ). Following the recommendations of [77] we discarded rooftop areas oriented towards north and

**Table 1**

Estimated uncertainties per technical model. \* Estimated, \*\* empirical model—estimated error of linearization, \*\*\* needs further validation.

| Model                           | rME (%)    | rRMSE (%) | RMSE (K)  | Typical value (%) | Acceptable value |
|---------------------------------|------------|-----------|-----------|-------------------|------------------|
| Building demand at urban scale  | 32 [1]     | –         | –         | 77 [1]            | 5%               |
| Roof-top solar                  | –          | 14 [119]  | –         | –                 | 30% [120]        |
| Geothermal heat                 | –          | –         | –         | –                 | 5%               |
| Lake water heat                 | –          | –         | 0.8 [121] | –                 | 1 K*             |
| Servers heat                    | 5 [1]**    | –         | –         | –                 | 5%               |
| Industrial processes***         | –          | –         | –         | –                 | 5%               |
| Sewage heat                     | –          | –         | 0.37 [82] | –                 | 1 K*             |
| Heat pumps                      | 6 [122]    | –         | –         | –                 | 5%               |
| Condensing boilers              | 2.5 [88]** | –         | –         | 2 [123]           | 5%**             |
| Combined gas cycle turbine      | 2 [124]    | –         | –         | –                 | 5%               |
| Fuel cell                       | 2 [89]**   | –         | –         | –                 | 5%*              |
| Photovoltaic panels             | –          | 1.8 [95]  | –         | –                 | 5%               |
| Solar collectors                | 5 [125]    | –         | –         | –                 | 5%               |
| Photovoltaic–thermal            | –          | 1.8 [95]  | –         | –                 | 5%*              |
| Chiller and cooling tower       | –          | 5 [101]   | –         | –                 | 5%               |
| Thermal storage***              | –          | –         | –         | –                 | 5%*              |
| Substations and heat exchangers | –          | –         | 1.2 [16]  | –                 | 1.2 K*           |
| Circulation pumps               | 2**        | –         | –         | –                 | 5%*              |

other rooftop areas with low solar potential (i.e. where  $I_{hor,k} < 0.75 I_{hor}^{max}$ ).

### 2.2.2. Ambient heat

**2.2.2.1. Geothermal heat.** The geothermal heat potential for an area  $k$  is determined in Eq. (1) considering the maximum thermal capacity per borehole ( $\phi_{BH}^{max}$ ), the separation between probes ( $S_{BH}$ ) and the maximum depth of exploration ( $Z_{BH}$ ).

$$\phi_{geo,th} = n_{BH} \cdot Z_{BH} \cdot \phi_{BH}^{max} \quad [Wh] \quad (1)$$

where

$$n_{BH} = \frac{\sqrt{A_k + S_{BH}}}{S_{BH}}$$

The time-dependent temperature of the source is calculated with Eq. (2) for depths up to 20 m. For higher depths, a linear increase of 0.03 C/m is assumed and the average linear value is then determined.

$$T_{gr} = \bar{T}_\alpha + (\bar{T}_\alpha^{max} - \bar{T}_\alpha) * e^{-\varepsilon} * \cos\left(2\pi \frac{t}{8760} - \varepsilon\right) + 273 \quad [K] \quad (2)$$

where

$$\varepsilon = Z^0 \sqrt{\frac{\pi \rho_g C_{pg}}{8760 k_g}}$$

**2.2.2.2. Lake water.** The resource potential of lake water is determined in Eq. (3). The maximum mass flow rates are given by Eq. (4) for a delta of temperature  $\Delta T$  (max /lake) [39]. The temperature variation at the intake is estimated in an hourly basis with the model of [78]. The model accounts for convective entrainment, wind-driven mixing, and volumetric solar radiation absorption. The model uses as input parameters hourly values of wind speed, air temperature, solar radiation, air humidity and cloudiness levels as well as characteristics of the lake related to depth and area. The later have been extracted from [79]. The governing equation of the model is presented in Eq. (5). Its solution method is outside of the scope of this article and can be found in [78].

$$\phi_{lake} = 0.27 \cdot \dot{V}_{lake} \cdot \rho_{surf} \cdot C_{pw} \cdot \Delta T_{lake}^{max} \quad [Wh] \quad (3)$$

$$\dot{m}_w^{max} = \frac{\phi_{lake}}{C_p \cdot (T_{w,z} - T_{DH/DC, re})} \quad [kg/s] \quad (4)$$

$$T_{W,z} = \begin{cases} T_{surf}, & 0 \leq z \leq h \\ T_{surf} - (T_{surf} - T_{bottom}) \cdot \phi_T, & x \geq 0 \end{cases} \quad [K] \quad (5)$$

where

$$\phi_T = \left(\frac{40}{3} C_{1,lake} - \frac{20}{3}\right) \cdot C_{2,lake} + (18 - 30 C_{1,lake}) \cdot C_{2,lake}^2 + (20 C_{1,lake} - 12) \cdot C_{2,lake}^3 + \left(\frac{5}{3} - \frac{10}{3} C_{1,lake}\right) \cdot C_{2,lake}^4;$$

$$C_{1,lake} = \frac{(0.3 \cdot 0.03 \cdot C_{4,lake}^2)}{(D_{lake} - h)^2} \cdot C_{3,lake};$$

$$C_{2,lake} = \frac{Z - h_{lake}}{D_{lake} - h_{lake}};$$

$$C_{3,lake} = \sqrt{-\frac{g \cdot \alpha T \cdot (T_z - T_{bottom})}{D_{lake} - h}};$$

$$C_{4,lake} = \sqrt{\frac{\rho \alpha}{\rho_0} C_{D,surf} \cdot v_{wind}^2}$$

### 2.2.3. Sources of waste heat

**2.2.3.1. Servers.** In Eq. (6) we normalized the empirical model of [80] to simulate the hourly excess heat from water-cooled data centers ( $\phi_{data,th}$ ) in function of the part load factor ( $\phi_{pl,data}$ ) and the internal temperature of servers ( $T_{data,re}$ ). The part load factor is function of the hourly electrical consumption of servers ( $\phi_{data,el}$ ) and their nominal power capacity ( $\phi_{data,el}^0$ ). Both quantities are obtained from the demand model of Section 2.1. Following the recommendations of [45] the internal temperature of water-cooled servers is maintained as high as 55 °C in order to obtain the maximum exergy to energy ratio efficiency. The mass flow rate of extraction of heat is considered for a constant delta of temperature  $\Delta T_{data}^0$ . For the analysis, we assumed the fan power included in the model to be equal to zero (i.e.  $\phi_{data,fan} = 0$ ). The costs of heat recovery from these units are estimated as two times the cost of heat exchange area as calculated in Section 2.3.7.

$$\phi_{H,gen,data} = \phi_{data,el} * n_{data} + \phi_{data,fan} \quad [Wh] \quad (6)$$

where

$$n_{data} = C_{1,data} + C_{2,data} \cdot \phi_{pl,data} + C_{3,data} \cdot T_{data,re} + C_{4,data} \cdot T_{data,re}^2;$$

$$\phi_{pl,data} = \frac{\phi_{data,el}}{\phi_{data,el}^0}$$



**2.2.3.2. Industrial processes.** We estimated the heat recovery potential of air compression for N<sub>2</sub> production ( $\phi_{\text{cair,th}}$ ) with the system of equations Eqs. (7)–(13). For this we used data about the hourly volume flow ( $V_{\text{ca,sup}}$ ) and pressure requirements of compressed air ( $P_{\text{ca,sup}}$ ) [1]. Based on the first law of thermodynamics we modeled a double stage compressor with a low pressure and a high-pressure cylinder. This widely used scheme allows for generating heat recovery opportunities at low (45–55 °C) and medium temperatures (65–90 °C) simultaneously. Finally, in Eq. (14), we calculated the total electricity consumption needed from the production of compressed air. Only costs of heat exchange area are included in Section 2.3.7.

$$P_{\text{ca,sup,LP}} = \sqrt{P_a \cdot P_{\text{ca,sup}}} \quad [\text{kPa}] \quad (7)$$

$$V_{\text{ca,sup}} = V_{\text{ca,sup,LP}} \left( \frac{P_{\text{ca,sup}}}{P_{\text{ca,sup,LP}}} \right)^{-1/n} \quad [\text{m}^3 \text{ h}] \quad (8)$$

$$V_a = V_{\text{ca,sup,LP}} \left( \frac{P_{\text{ca,sup,LP}}}{P_a} \right)^{-1/n} \quad [\text{m}^3 \text{ h}] \quad (9)$$

$$T_{\text{ca,sup,LP}} = T_a \left( \frac{P_{\text{ca,sup,LP}}}{P_a} \right)^{(n_{\text{a,pol}}-1)/n_{\text{a,pol}}} \quad [\text{K}] \quad (10)$$

$$T_{\text{ca,sup,HP}} = T_{\text{ca,sup,LP}} \left( \frac{P_{\text{ca,sup}}}{P_{\text{ca,sup,LP}}} \right)^{(n_{\text{a,pol}}-1)/n_{\text{a,pol}}} \quad [\text{K}] \quad (11)$$

$$\begin{aligned} \phi_{H,\text{gen,ca,HP}} &= \frac{n P_{\text{ca,sup,LP}} V_{\text{ca,sup,LP}}}{n_{\text{a,pol}} - 1} \left[ \left( \frac{P_{\text{ca,sup}}}{P_{\text{ca,sup,LP}}} \right)^{(n_{\text{a,pol}}-1)/n_{\text{a,pol}}} - 1 \right] \quad [\text{Wh}] \end{aligned} \quad (12)$$

$$\begin{aligned} \phi_{H,\text{gen,ca,LP}} &= \frac{n_{\text{a,pol}} P_{\text{air}} V_{\text{air}}}{n_{\text{a,pol}} - 1} \left[ \left( \frac{P_{\text{cair,sup,LP}}}{P_{\text{air}}} \right)^{(n_{\text{a,pol}}-1)/n_{\text{a,pol}}} - 1 \right] \quad [\text{Wh}] \end{aligned} \quad (13)$$

$$\begin{aligned} \phi_{E,\text{ca}} &= \frac{n_{\text{a,pol}} P_a V_a}{n_{\text{a,pol}} - 1} \cdot \left[ \left( \frac{P_{\text{ca,sup}}}{P_a} \right)^{(n_{\text{a,pol}}-1)/n_{\text{a,pol}}} - 1 \right] \\ &\cdot (n_{\text{exp}} \cdot n_{\text{engine}})^{-1} \quad [\text{Wh}] \end{aligned} \quad (14)$$

**2.2.3.3. Sewage.** We evaluated both volume flows and temperatures of sewage lines influenced by both buildings on site and upstream recollection areas at the district level. For buildings on site and those of surrounding recollection areas, we calculated the domestic hot water and fresh water requirements with the demand model of Section 2.1 and estimated the combined outflow per building. For this we assumed a complete mix of both outflows, a ratio of wastewater/fresh water intake of 95%, and thermal losses in building's internal sewage lines given by Eq. (15). In this equation mass flow rates of domestic hot water ( $\dot{m}_{\text{ww}}$ ), fresh water ( $\dot{m}_{\text{fw}}$ ), length of piping ( $L_{\text{sw}}$ ) and internal building temperature ( $T_{a,b}$ ) are obtained directly from the demand model.

$$\phi_{\text{dis,l,nr,sw}} = (T_{\text{spur}} - T_x) \cdot \dot{m}_{\text{sw}} C_{p_w} \quad [\text{Wh}] \quad (15)$$

where

$$T_x = T_{a,b} + (T_{\text{spur}} - T_{a,b}) \cdot e;$$

$$T_{\text{spur}} = \frac{0.95 \cdot (T_{\text{ww}} \cdot \dot{m}_{\text{ww}} + T_{\text{fw}} \cdot \dot{m}_{\text{fw}})}{\dot{m}_{\text{sw}}};$$

$$\dot{m}_{\text{sw}} = 0.95 \cdot (\dot{m}_{\text{ww}} + \dot{m}_{\text{fw}})$$

$$\phi_{\text{dis,l,nr,sw}} = (T_{\text{spur}} - T_{\text{sw}}) \cdot \dot{m}_{\text{sw}} C_{p_w} \quad [\text{Wh}] \quad (16)$$

The resulting temperature at the exit of every building is calculated by solving Eq. (16) for  $T_{\text{sw}}$ . We determined the location of a pipe section of at least 100 m long and 0.8 m of diameter with the help of GIS data<sup>1</sup>. Then, we determined the length of the sewage network ( $L_{\text{sw,net}}$ ) connecting all costumers to this point. In order to account for wastewater inflows beyond the site, we gathered GIS data of regional sewage lines and identified the accumulated flow passing through every recollection area. For this we determined a normalized profile of mass flow rates and temperatures for all buildings in each recollection area and assumed it to be representative of that coming from regional lines. The normalized profile is multiplied for an average regional peak consumption of 196 l.p.d [81]. For this the number of population connected to the regional lines is directly obtained from GIS data. A temperature decay is calculated in regards to distance to the site as 4 °C/km [82]. For this we assume a minimum temperature in the sewage of that of the soil. We then estimated a final regional temperature ( $T_{\text{sw,region}}$ ) and mass flow rate ( $\dot{m}_{\text{sw,region}}$ ). With these values we computed a temperature and mass flow rate of the sewage network including all buildings in the area and assuming complete mixing in the following equations:

$$\dot{m}_{\text{sw,all}} = \sum \dot{m}_{\text{sw}} + \dot{m}_{\text{sw,region}} \quad [\text{kg/s}] \quad (17)$$

$$T_{\text{sw,all}} = \frac{\sum \dot{m}_{\text{sw}} \cdot T_{\text{sw}} + \dot{m}_{\text{sw,region}} \cdot T_{\text{sw,region}}}{\dot{m}_{\text{sw,all}}} \quad [\text{kg/s}] \quad (18)$$

Following the approach of [82] we calculated the losses of the sewage network at this point in Eq. (19). The approach considers losses driven by temperatures in the sewage water ( $T_{\text{sw}}$ ) the surrounding ground ( $T_g$ ), the in-sewer air ( $T_{\text{sw,a}}$ ) and overall thermal resistance factors between wastewater and in-sewer air (Eq. (20)) and between waste water and soil (Eq. (21)). In these equations, both the wet perimeter (wet.p) and the thickness of the wastewater flow layer ( $b_{\text{sw,lyr}}$ ) are a function of the mass flow rate, the velocity of the flow ( $\dot{v}_{\text{sw,all}}$ ) and the diameter of the sewer ( $D_{\text{sw}}$ ). The solution to these equations is outside of the scope of this article and can be found in [82].

$$\begin{aligned} \phi_{\text{dis,l,nr,sw,all}} &= (T_{\text{sw,all}} - T_{\text{sw,a}}) \cdot \frac{1}{R_{\text{sw,a}}} + (T_{\text{sw,all}} - T_g) \cdot \frac{1}{R_{\text{sw,g}}} \quad [\text{Wh}] \end{aligned} \quad (19)$$

$$R_{\text{sw,a}} = \frac{b_{\text{sw,p}}}{5.85 \cdot \sqrt{\dot{v}_{\text{sw,a}} \cdot b_{\text{sw,lyr}} \cdot L_{\text{sw,net}}}} \quad [\text{Wh}] \quad (20)$$

where

$$b_{\text{sw,lyr}} = f(\dot{m}_{\text{sw,all}}, \dot{v}_{\text{sw,all}}, D_{\text{sw}})$$

$$R_{\text{sw,g}} = \frac{b_{\text{sw,p}}}{K_{\text{sw,p}} \cdot \text{wet.p} \cdot L_{\text{sw,net}}} + \frac{Z_g}{K_g \cdot \text{wet.p} \cdot L_{\text{sw,net}}} \quad [\text{Wh}] \quad (21)$$

where

$$\text{wet.p} = f(\dot{m}_{\text{sw,all}}, \dot{v}_{\text{sw,all}}, D_{\text{sw}})$$

<sup>1</sup> These are the minimum dimensions of a pipe section for the use of the heat extraction technology in the sewage bed.

**Table 2**  
Techno-economic boundary conditions per technology.

| Technology                      | Lt [yr] | Ir [%] | Min Size [MW] | Max Size [MW] | Min $\phi_{pl}$ [%] |
|---------------------------------|---------|--------|---------------|---------------|---------------------|
| Heat pump                       | 20      | 5      | 0.001 th      | 20 th         | 20                  |
| Condensing boiler               | 20      |        | 0.001 th      | 210 th        | 10                  |
| Combined cycle gas turbine      | 25      | 6      | 20 el         | 50 el         | 40                  |
| Fuel cells                      | 10      | 5      | 0.001 th      | 0.005 th      | 20                  |
| Vapor compression chiller       | 20      | 5      | 0.001 th      | 20 th         | 10                  |
| Seasonal thermal storage        | 60      | 5      | –             | –             | –                   |
| Thermal distribution networks   | 60      | 4      | –             | –             | 20                  |
| Substations and heat exchangers | 20      | 4      | –             | –             | –                   |
| Circulation pumps               | 10      | 4      | 0.001 th      | 20 th         | 20                  |
| Solar technologies              | 20      | 5      | –             | –             | 20                  |

**Table 3**  
Performance factors for different technologies.

| Technology                                                   | PF <sub>EP</sub> [MJ/MJ] | PF <sub>GHE</sub> [CO <sub>2</sub> -eq kg/MJ] |
|--------------------------------------------------------------|--------------------------|-----------------------------------------------|
| Natural gas—fired condensing boiler                          | 1.403                    | 0.100                                         |
| Natural gas—fired fuel cell                                  | 1.356                    | 0.097                                         |
| District cooling—vapor compression chiller and cooling tower | 1.19                     | 0.0282                                        |
| District heating—Natural gas-fired boiler                    | 1.550                    | 0.087                                         |
| District heating—Biogas-fired boiler                         | 0.512                    | 0.0125                                        |
| District heating—Natural gas-fired combined cycle            | 0.641                    | 0.0377                                        |
| District heating—Biogas-fired combined cycle                 | 0.212                    | 0.0192                                        |
| District cooling—Lake water/water heat pump                  | 0.897                    | 0.0153                                        |
| District cooling—Lake water pump                             | 0.155                    | 0.002                                         |
| District heating—Lake water/water heat pump                  | 0.897                    | 0.015                                         |
| Local grid electricity                                       | 2.63                     | 0.0413                                        |
| Photovoltaic panels                                          | 0.278                    | 0.0202                                        |
| Photovoltaic-thermal panels—electricity                      | 0.222                    | 0.0108                                        |
| Solar collectors                                             | 0.201                    | 0.0091                                        |
| Ground source heatpump—water/water                           | 0.695                    | 0.0164                                        |
| Water source heatpump—water/water                            | 0.795                    | 0.0179                                        |
| Sewage source heatpump—water/water                           | 0.795                    | 0.0179                                        |
| Data center source heatpump—water/water                      | 0.795                    | 0.0179                                        |
| Processes source heatpump—water/water                        | 0.795                    | 0.0179                                        |

On the other hand, the final temperature upstream of the selected pipeline is calculated after losses in the following equation:

$$T_{sw} = \frac{(T_{sw,all} - T_{sw,a}) \cdot 1/R_{wa} + (T_{sw,all} - T_g) \cdot 1/R_{sw,g}}{\dot{m}_{sw,all} C_{p,sw}} \quad [\text{Wh}] \quad (22)$$

### 2.3. Systems technology module

The systems technology module aggregates techno-economic models of several production, storage and energy distribution units. In Table 2 [48] and Table 3 [83], we present techno-economic boundary conditions and performance factors per technology. The estimated accuracy of each model as well as its typical and acceptable values are presented in Table 1.

#### 2.3.1. Heat pumps

For a certain thermal load  $\phi_{th}$ , the compression power of a heat pump is calculated with Eq. (23). We consider that the compression power represents 83% of the total electrical needs [84]. In

consequence the extra auxiliary energy of circulation pumps is calculated in Eq. (24). The coefficient of performance of water source (WSHP) and ground source (GSHP) heat pumps is described in Eq. (25) [85]. It is a function of the exergetic efficiency  $\eta_{ex}$  (0.6 for WSHP [4] and 0.67 for GSHP [86]), the temperature of the source  $T_{source}$ , the temperature of the sink  $T_{sink}$  and the approach temperature at the condenser and evaporator  $\Delta T_{A,HP}$ .

$$\phi_{E,HP} = \frac{\phi_{H,gen,HP}}{COP_{HP}} \quad [\text{Wh}] \quad (23)$$

$$\phi_{E,aux,HP} = 0.17 \cdot \phi_{E,HP} \quad [\text{Wh}] \quad (24)$$

$$COP_{HP} = \eta_{ex} \cdot \left( 1 - \frac{T_{source} - \Delta T_{A,HP}/2}{T_{sink} + \Delta T_{A,HP}/2} \right)^{-1} \quad [-] \quad (25)$$

The investment costs for WSHPs are calculated according to the nominal thermal capacity of the unit ( $\phi_{th}^0$ ) in Eq. (26) [85]. For every time span, the thermal output of the unit and its nominal value are restricted to the maximum amount of heat that can be obtained from the source in Eq. (27).

$$InvC_{WSHP} = \left( -493.53 \ln \frac{\phi_{H,gen,HP}^0}{1000} + 5484 \right) \cdot \frac{\phi_{H,gen,HP}^0}{1000} \cdot \phi_{ann} \quad [\text{EU}] \quad (26)$$

where

$$\phi_{ann} = \frac{Ir \cdot (1 + Ir)^{Lt}}{(1 + Ir)^{Lt} - 1}$$

$$\phi_{H,gen,th} \leq \phi_{source,th} \cdot \left( 1 - \frac{1}{COP_{HP}} \right)^{-1} \quad [\text{Wh}] \quad (27)$$

In the case of GSHPs, the investment costs include those of every borehole in Eq. (28) [87].

$$INV_{GSHP} = \left[ 6297 (\phi_{H,gen,HP}^0)^{0.49} + 8520 (\phi_{H,gen,HP}^0)^{0.74} \right] \cdot \phi_{ann} \quad [\text{EU}] \quad (28)$$

O&M costs are derived from the resulting electrical needs and electricity prices ( $\xi_{el,CH}$ ). Maintenance costs rated at 1% of the total investment costs.

O&M<sub>GSHP</sub>

$$= 0.01 \cdot InvC_{HP} + \sum \frac{(\phi_{E,HP} + \phi_{E,aux,HP}) \cdot \xi_{el,CH}}{1000} \quad [\text{EU}] \quad (29)$$

#### 2.3.2. Condensing boilers

Condensing—modulating boilers are considered to burn both biogas and natural gas at their high heat value (HHV). For this we used the empirical model described in Eqs. (30)–(32) [88] based on

supply and return temperatures customer side and the part load efficiency ( $n_{\phi_{pl},boiler}$ ).

$$\phi_{fuel,u,boiler} = \frac{\phi_{H,gen,boiler}}{n_{gen,boiler}} \quad [Wh] \quad (30)$$

$$n_{gen,boiler} = n_{\phi_{pl},boiler} \cdot \left[ \frac{(T_{sup} + T_{re}/2) - T_a}{\Delta T_{test}} \right]^{1.25} \quad [-] \quad (31)$$

$$n_{\phi_{pl},boiler} = C_{1,boiler} + C_{2,boiler} \cdot \log(\phi_{pl,boiler}) \quad [-] \quad (32)$$

where

$$\phi_{pl,boiler} = \frac{\phi_{th}}{\phi_{th}^0}$$

The investment costs are given in Eq. (33). Operational costs are derived from the fuel consumption of the unit and fuel prices ( $P_{NG,CH}$ ,  $P_{BG,CH}$ ) and maintenance costs are rated at 1% of total investment costs in Eq. (34).

$$INVC_{boiler} = \begin{cases} 8400 + 0.014 \cdot \phi_{H,gen,boiler}^0 \cdot \phi_{H,gen,boiler} & \phi_{H,gen,boiler} > 320,000 \text{ W} \\ 0.8 \cdot \phi_{H,gen,boiler}^0 \cdot \phi_{H,gen,boiler} & \phi_{H,gen,boiler} \leq 320,000 \text{ W} \end{cases} \cdot \phi_{ann} \quad [EU] \quad (33)$$

$$O\&MC_{boiler} = 0.01 \cdot INVC_{HP} + \sum \frac{\phi_{fuel,u} \cdot \xi_{NG \text{ or } BG,CH}}{1000} \quad [EU] \quad (34)$$

### 2.3.3. Combined heat and power technologies

We modeled one central and one decentralized combined heat and power (CHP) unit both assisted by either natural gas or biogas. The first consists in a combined gas cycle turbine; the second consists in a solid-oxide fuel cell.

**2.3.3.1. Combined gas cycle turbine.** Following the approach of [85] we modeled a combined gas cycle turbine with heat recovery. The fuel consumption of the unit is described in the following equation:

$$\phi_{fuel,u,cc} = \frac{\phi_{E,gen,cc}}{\eta_{cc,el}} \quad [Wh] \quad (35)$$

The electrical efficiency of the unit is dependent on the electrical part load of the unit, presented in the following equation [85]:

$$\eta_{cc} = (0.4089 + 0.9624\phi_{pl,cc} - 0.3726\phi_{pl,cc}^2) \cdot \eta_{gas,el}^0 \quad [-] \quad (36)$$

where

$$\eta_{gas,el}^0 = 0.0196 \ln \phi_{E,gen,cc}^0 + 0.1317;$$

$$\phi_{pl,cc} = \frac{\phi_{E,gen,cc}}{\phi_{E,gen,cc}^0}$$

On the other hand, the thermal output of the unit in Eq. (37) is constrained to the temperature of condensation of a potential district heating network ( $T_{cond}$ ) and the low and high-pressure mass flow rates of the double pressure steam turbine ( $\dot{m}_{LP}$ ,  $\dot{m}_{HP}$ ) given in Eq. (38). The solution to Eq. (38) is outside of the scope of this article but can be found in [85] for both natural and biogas as fuels.

$$\phi_{H,gen,cc} = (\dot{m}_{LP} + \dot{m}_{HP}) \cdot [-2.4967(T_{cond}) + 2507] \quad [Wh] \quad (37)$$

where

$$T_{cond} = \frac{T_{DH,sup} + T_{DH,re}}{2} + \Delta T_{cond}$$

$$\dot{m}_{LP}, \dot{m}_{HP} = f(\eta_{cc,el}, \dot{m}_{exh}, \Delta T_{HRSG}, T_{HP}, T_{LP}) \quad [kg/s] \quad (38)$$

The overall electricity production of the unit is given in Eq. (39). In this equation, the gross electricity production of the steam turbine ( $\phi_{el,turb}$ ) and the electricity consumed by the steam compressor ( $\phi_{el,comp}$ ) are presented in Eqs. (40) and (41).

$$\phi_{E,gen,cc} = \eta_{cc,el} \cdot (\phi_{el,turb} - \phi_{el,comp}) \quad [Wh] \quad (39)$$

$$\dot{E}_{st}^{comp} = \nu_w (\dot{m}_{LP}(P_{LP} - P_{cond}) + (\dot{m}_{LP} + \dot{m}_{HP})(P_{HP} - P_{HP})) \quad [Wh] \quad (40)$$

where

$$P_x = 0.0261 T_x^2 - 2.1394 T_x + 52.893; \quad \forall x \in (HP, LP, cond)$$

$$\dot{E}_{st}^{turb} = \dot{m}_{HP} (C_{1,st} - C_{2,st}) + (\dot{m}_{LP} + \dot{m}_{HP}) (C_{2,st} - C_{3,st}) \quad [-] \quad (41)$$

where

$$C_{1,st} = 2.5081(T_{exh} - \Delta T_{HRSG}) + 2122.7;$$

$$C_{2,st} = 2.3153 T_{inter} + 2314.7;$$

$$C_{3,st} = 1.6979 T_{cond} + 2506.6;$$

$$T_{exh} = (0.7379 + 0.2621\phi_{pl,st}) \cdot \begin{cases} 1053 \text{ K for biogas} \\ 956 \text{ K for natural gas} \end{cases}$$

The investment costs are computed with Eq. (42) [85]. O&M costs are derived from the fuel needs of the unit and a maintenance cost depending on the electrical demand attended in Eq. (43).

$$INVC_{CC} = 32978 \cdot \frac{\phi_{el}^0}{1000}^{0.5967} \cdot \phi_{ann} \quad [EU] \quad (42)$$

$$O\&MC_{CC} = 0.03 \cdot \sum \frac{\phi_{el}}{1000} + \sum \frac{\phi_{fuel} \cdot \xi_{NG \text{ or } BG,CH}}{1000} \quad [EU] \quad (43)$$

**2.3.3.2. Fuel cells.** The thermal and electrical output of a solid oxide fuel cell is defined from the empirical model of [89] in Eqs. (44)–(46).

$$\phi_{fuel,u,FC} = \frac{\phi_{gen,FC,th}}{n_{FC,th}} + \frac{\phi_{gen,FC,el}}{n_{FC,el}} \quad [Wh] \quad (44)$$

$$n_{FC,el} = n_{FC,el}^0 \cdot [-0.220 + 5.277\phi_{pl,FC} - 9.127\phi_{pl,FC}^2 + 7.172\phi_{pl,FC}^3 - 2.103\phi_{pl,FC}^4] \quad [Wh] \quad (45)$$

where

$$\phi_{pl,FC} = \frac{\phi_{gen,FC,el}}{\phi_{gen,FC,el}^0}$$

$$n_{FC,th} = n_{FC,th}^0 \cdot [0.9 - 0.07\phi_{pl,FC} + 0.17\phi_{pl,FC}^2] \quad [Wh] \quad (46)$$

The investment costs are computed with Eq. (47) [90]. O&M costs are derived of fuel needs and electricity production in Eq. (48) [90].

$$INVC_{CC} = 12000 \cdot \frac{\phi_{th}^0}{1000} \cdot \phi_{ann} \quad [EU] \quad (47)$$

$$O\&MC_{CC} = 0.05 \cdot INVC_{CC} + \sum \frac{\phi_{fuel} \cdot \xi_{NG \text{ or } BG,CH}}{1000} \quad [EU] \quad (48)$$



### 2.3.4. Solar technologies

We estimated the technical potential of rooftop solar collectors (SC), photovoltaic (PV) and photovoltaic thermal (PV/T) panels at their optimal orientation ( $\theta_m^0$ ), slope ( $B_m^0$ ) and spacing ( $S_m^0$ ). For flat roofs (slope  $< 5^\circ$ ), we assumed fixed and south oriented modules and estimated  $B_m^0$  according to Eq. (49) [91], and  $S_m^0$  according to Eq. (50). For other cases, both  $\theta_m^0$  and  $B_m^0$  were assumed to follow the orientation and slope of the roof and  $S_m^0$  to equal zero.

$$B_m^0 = \cos(\theta_m^0) \cdot \tan(L_m) \cdot \left( \frac{1}{1 + (\tau\alpha)_D \cdot C_{1,sol} - (\tau\alpha)_G \cdot R_{flg}/2 \cdot (1 - C_{1,sol})} \right) [^\circ] \quad (49)$$

where

$$C_{1,sol} = \begin{cases} 0.977, & C_{2,sol} \leq 0.15 \\ 1.23 - 1.361C_{2,sol}, & 0.15 < C_{2,sol} \leq 0.70 \\ 0.273, & C_{2,sol} > 0.70 \end{cases}$$

$$As_m = \frac{w_m \cdot \sin(B_m) \cdot \cos(180 - \theta_m^0)}{\tan(h_{sol})} [m] \quad (50)$$

We grouped areas with similar values of  $I_{hor,yr}$ ,  $B_m^0$  and  $\theta_m^0$  with a non-spatially constrained use of the k-means clustering algorithm of ArcGIS 10.3. Based on the approach of [1] the optimal number of groups ( $N_{gr}$ ) was evaluated using the Calinski–Harabasz pseudo F-statistic [92].

The potential for each solar technology is hereafter evaluated for each group to later on be aggregated both at the level of costs and energy outputs.

**2.3.4.1. Photovoltaic panels.** PV production is calculated according to the empirical correlation of Eq. (51) [93] taking into account only surfaces with solar isolation above  $50 \text{ kWh/m}^2$  [94]. Eq. (51) expresses the variation of power production as a function of cell temperature (Eq. (52)) [95], the PV panel's aperture area (Eq. (53)), the absorbed solar radiation (Eq. (54)) [93], the expected system losses ( $\phi_{sys,l,pv}$ ), and the technical properties of the Annex B.

$$\phi_{E,gen,pv} = n_{pv}^0 \cdot A_{ap,pv} \cdot I_{ab,pv} [1 - \beta_{ref} \cdot (T_{pv} - T_{pv}^0)] \cdot (1 - \phi_{sys,l,pv}) [Wh] \quad (51)$$

$$T_{pv} = T_a + I_{ab,pv} \cdot \left( \frac{T_{pv}^0 - 293}{800} \right) [K] \quad (52)$$

$$A_{ap,pv} = A_{ap,sc} = A_{ap,pv/T} = ind_{gr} \cdot L_{mesh} \cdot \left( \frac{L_{mesh} - S_{pv}}{\cos(B_m^0)} \right) [m^2] \quad (53)$$

$$I_{ab,pv} = M \left( I_{sol,B}(\tau\alpha)_B + I_{sol,D}(\tau\alpha)_D \left[ \frac{1 + \cos(B_m^0)}{2} \right] + I_{sol,g}(\tau\alpha)_g \left[ \frac{1 - \cos(B_m^0)}{2} \right] \right) [W/m^2] \quad (54)$$

In Eq. (53)  $ind_{gr}$  represents the number of individual points contained in every group. The absorbed solar radiation is a function of the air mass ( $M$ ), which is derived using the characteristics of a mono-crystalline PV-module described in detail in [96]. On

the other hand, the spectral components of solar isolation (i.e.  $I_{sol,B}(\tau\alpha)_B$ ,  $I_{sol,D}(\tau\alpha)_D$ , and  $I_{sol,g}(\tau\alpha)_g$ ) are evaluated according to [97].

In terms of economics, investment costs are described as a function of the peak power capacity installed in every rooftop (Eq. (55)) in Eq. (56). O&M costs include feed-in-tariffs and local remuneration schemes per building (Eq. (57)).

$$PV_{pk,b} = n_{pv}^0 \cdot \sum_{j=1}^{N_b} \sum_{i=1}^{ind_{gr,b}} \left( \frac{A_{ap,pv}}{ind_{gr}} \right)_{i,j} [kWp] \quad (55)$$

$$\forall i \in ind_{gr,b}; \forall j \in N_b$$

$$INVC_{pv} = \sum_{i=1}^{N_b} INVC_{pv,b_i} \cdot \varphi_{ann} [EU] \quad (56)$$

where

$$INVC_{pv,b} = \begin{cases} 3500 PV_{pk,b} & \text{if } PV_{pk,b} < 10 \text{ kW p} \\ 2500 PV_{pk,b} & \text{if } 10 \text{ kW} < PV_{pk,b} < 200 \text{ kW p} \end{cases} [EU]$$

$$O\&M_{pv} = \sum_{i=1}^{N_b} O\&M_{pv,b_i} \quad (57)$$

where

$$O\&M_{pv,b} = 0.01 \cdot INVC_{pv,b} - \begin{cases} -(0.00645 \ln(PV_{pk,b}) + 0.212) \phi_{E,gen,pv} & \text{if } PV_{pk,b} < 200 \text{ kW} \\ 0.176 \phi_{E,gen,pv} & \text{if } 200 \text{ kW} \leq PV_{pk,b} \end{cases}$$

**2.3.4.2. Solar collectors.** The heat production of SC is calculated according to an adaptation of the quasi-dynamic model of [97,98]. The resulting Eq. (58) expresses the variation of heat production as a function of collector's and ambient temperature Eq. (59), the absorbed solar radiation in Eq. (60) [97], and the technical properties of Annexes A and B.

$$\phi_{H,gen,sc} = A_{ap,sc} (n_{sc}^0 \cdot I_{ab,sc} - C_{1,sc} \cdot \Delta T_{sc} - C_{2,sc} |\Delta T_{sc}| \cdot \Delta T_{sc} - C_{eff,sc} \frac{\Delta T_{sc}}{\Delta t}) - \phi_{dis,l,nr,sc} [Wh] \quad (58)$$

$$\Delta T_{sc} = \frac{T_{sc,sup} + T_{sc,re}}{2} - T_a [K] \quad (59)$$

$$I_{ab,sc} = I_{sol,B}(\tau\alpha)_B + I_{sol,D}(\tau\alpha)_D \left[ \frac{1 + \cos(B_m^0)}{2} \right] + I_{sol,g}(\tau\alpha)_g \left[ \frac{1 - \cos(B_m^0)}{2} \right] [W/m^2] \quad (60)$$

In Eq. (58) we included two terms to represent non-recoverable ( $\phi_{dis,l,nr,sc}$ ) and recoverable ( $\phi_{dis,l,r,sc}$ ) losses from the source to the heat exchanger in the local thermal grid. The first is calculated according to Eq. (61), the second is equivalent to 50% of the electricity required by pumps in the circulation loop in Eq. (62) [99].

$$\phi_{dis,l,nr,sc} = \psi_{sc} \cdot L_{sc,p,ext} \cdot A_{ap,sc} \cdot \Delta T_{sc} \cdot \frac{\dot{m}_{sc}}{\dot{m}_{sc}^0} - 0.5 \phi_{E,aux,sc} [Wh] \quad (61)$$

where

$$L_{sc,p,ext} = \frac{2 \cdot L_{mesh} \cdot ind_{gr}}{A_{ap,sc} \cdot A_{ap,sc} / A_{m,sc}}$$

$$\phi_{E,aux,sc} = \frac{\dot{m}_{sc} \cdot (\Delta P_{sc} + \Delta P_{L_{sc,p}} \cdot L_{sc,p} \cdot A_{ap,sc})}{\rho_{w/gl}} \cdot \left( \frac{\dot{m}_{sc}}{\dot{m}_{sc}^0} \right)^3; \quad [Wh] \quad (62)$$

$$\forall \Delta P_{sc} \in \{ \Delta P_{sc}^0, \Delta P_{sc}^{\min}, \Delta P_{sc}^{\max}, 0 \};$$

$$\forall \dot{m}_{sc} \in \{ \dot{m}_{sc}^0, \dot{m}_{sc}^{\min}, \dot{m}_{sc}^{\max}, 0 \}$$

where

$$L_{sc,p} = L_{sc,p,int} + L_{sc,p,ext};$$

$$L_{sc,p,int} = \frac{2 \cdot \sum_{N_b}^{i=1} h_{b,i}}{A_{ap,sc} \cdot A_{ap,sc} / A_{m,sc}}$$

The solution to Eq. (58) is an iterative process, which relates the solar collector's thermal output to its mass flow rate, Eq. (63). In order to increase this thermal output we derived from a combinatorial approach the mass flow rate that maximizes the energy generation function of Eq. (63) in every time-step. In this equation, the electricity required for pumping ( $\phi_{E,aux,sc}$ ) is assumed as twice as valuable as the thermal output of the solar collectors ( $\phi_{H,gen,sc}$ )<sup>2</sup>. For the sake of simplicity, the problem is subjected to four possible flow conditions ( $\dot{m}_{sc}^0, \dot{m}_{sc}^{\min}, \dot{m}_{sc}^{\max}, 0$ ) and pressure drops (i.e.  $\Delta P_{sc}^0, \Delta P_{sc}^{\min}, \Delta P_{sc}^{\max}, 0$ )

$$\phi_{H,gen,sc} = \dot{m}_{sc} C_{p,w/gl} \cdot (T_{sc,sup} - T_{sc,re}) - 2 \cdot \phi_{E,aux,sc} \quad [Wh] \quad (63)$$

The calculation process is carried out for every group in the area and the results are aggregated. For this, we assumed all modules to be connected in parallel and operate at a target temperature of 10 degrees above the return line of the community wise micro-grid.

Investment costs are estimated according to the collector's area in Eq. (64). O&M are given as a percentage of the investment costs in Eq. (65).

$$INV_{SC} = 2050 \cdot A_{ap,sc} \cdot \varphi_{ann} \quad [EU] \quad (64)$$

$$O\&M_{SC} = 0.05 \cdot INV_{SC} \quad [EU] \quad (65)$$

**2.3.4.3. Photovoltaic-thermal panels.** Following the recommendations of [97,100], we estimated the electrical and thermal output of PV-T panels according to Eqs. (66) and (67). These equations account for the variability of the PV cell temperature in relation to the temperature of the circulating fluid and its positive effects on the reduction of thermal and electrical losses.

$$\begin{aligned} \phi_{E,gen,pv/T} &= \phi_{E,gen,pv} \\ &= f \left( n_{pv}^0, A_{ap,pv}, I_{ab,pv}, \beta_{ref}, T_{pv}, T_{pv}^0, \phi_{sys,l,pv} \right); \quad [Wh] \end{aligned} \quad (66)$$

$$\begin{aligned} \forall T_{pv} &= \left\{ \frac{T_{sc,sup} + T_{sc,re}}{2} \right\} \\ \phi_{H,gen,pv} &= \phi_{E,gen,pv} \\ &= f \left( A_{ap,pv}, n_{sc}^0, I_{ab,sc}, C_{1,sc}, \Delta T_{sc}, C_{2,sc}, C_{eff,SC}, \phi_{dis,l,nr,sc} \right) \quad [Wh] \end{aligned} \quad (67)$$

$$\forall C_{1,sc} = (C_{1,sc} - n_{pv}^0 \cdot \beta_{ref} \cdot I_{ab,pv});$$

$$\forall n_{sc}^0 = 1 - (n_{pv}^0 (1 - \beta_{ref}) \cdot (T_{pv} - T_{pv}^0))$$

For the sake of simplicity we analyzed a PV-T panel with similar properties of mono-crystalline cells and flat-plate collectors.

Investment and O&M costs are calculated according to Eqs. (68) and (69).

$$INV_{C_{pv/T}} = \sum_{i=1}^{N_b} INV_{C_{pv,T,b_i}} \quad [EU] \quad (68)$$

where

$$INV_{C_{pv/T,b}} = 5000 \cdot PV_{pk,b_i} \cdot \varphi_{ann}$$

$$O\&M_{C_{pv/T}} = \sum_{i=1}^{N_b} O\&M_{C_{pv,T,b_i}} \quad [EU] \quad (69)$$

where

$$\begin{aligned} O\&M_{C_{pv,b}} &= 0.01 \cdot INV_{C_{pv/T,b}} \\ &- \begin{cases} - (0.00645 \ln(PV_{pk,b}) + 0.212) \phi_{E,gen,pv/T} & \text{if } PV_{pk,b} < 200 \text{ kW} \\ 0.176 \phi_{E,gen,pv/T} & \text{if } 200 \text{ kW} \leq PV_{pk,b} \end{cases} \end{aligned}$$

### 2.3.5. Chillers and cooling tower

The model for vapor compression chillers uses the Gordon-Ng approach [101] to represent the four-stage thermodynamic cycle, where a refrigerant extracts heat for the cooling service at the condenser and transfers it to fresh water at the evaporator. The corresponding coefficient of performance is calculated in the following equation:

$$\begin{aligned} \frac{T_{w,re}}{T_{w,sup}} \left( 1 + \frac{1}{COP_{vcc}} \right) - 1 \\ = C_{1,vcc} \frac{T_{w,re}}{\phi_{C,gen,vcc}} + C_{2,vcc} \frac{T_{w,sup} - T_{w,re}}{T_{w,sup} \phi_{C,gen,vcc}} \\ + C_{3,vcc} \frac{\phi_{C,gen,vcc}}{T_{w,sup}} \left( 1 + \frac{1}{COP_{vcc}} \right) \quad [-] \end{aligned} \quad (70)$$

The compression work of the chiller needed to attend a cooling load  $\phi_{C,gen,vcc}$  is presented in the following equation:

$$\phi_{E,vcc} = \frac{\phi_{C,gen,vcc}}{COP_{vcc}} \quad [Wh] \quad (71)$$

A wet cooling tower is modeled downstream from the chiller. Its performance depends on the nominal cooling capacity and the cooling part load factor, both values are described in Eq. (72) [102].

<sup>2</sup> This assumption is based on a typical ratio of electricity and heat production out of primary energy of 2:1 [126].

$$\frac{\phi_{E,wct}}{\phi_{E,wct}^0} = 0.8603 \phi_{pl,vcc}^3 + 0.2045 \phi_{pl,vcc}^2 - 0.0623 \phi_{pl,vcc} + 0.0026 \quad [-] \quad (72)$$

where

$$\phi_{E,wct}^0 = 0.011 \phi_{C,gen,vcc}^0$$

$$\phi_{pl,vcc} = \frac{\phi_{C,gen,vcc}}{\phi_{C,gen,vcc}^0}$$

The investment costs for the two units is presented in Eq. (73). O&M costs are derived from the electricity needs plus 5% costs over the investment costs in Eq. (74).

$$INVC_{CT} = (386 \phi_{C,gen,vcc}^0 + 15.4 \phi_{C,gen,vcc}^0 + 1400) \cdot \varphi_{ann} \quad [EU] \quad (73)$$

$$O\&MC_{CHCT} = 0.05 \cdot InvC_{HP} + \sum \frac{(\phi_{E,wct} + \phi_{E,vcc}) \cdot \xi_{el,CH}}{1000} \quad [EU] \quad (74)$$

### 2.3.6. Thermal storage

Following the approach of [103,104] we modeled a daily and a seasonal thermal storage both consisting of a cylindrical water tank operating at temperatures between 20 and 90 °C. The daily storage tank is dedicated to store any thermal load; whereas the seasonal tank is dedicated to store only solar energy produced in site. Both storages are operated so the energy content at the end of the period of concern is the same as the initial. The thermal balance of both storage units is presented in Eq. (75). In this equation  $\phi_{H,gen}$  is the energy injected to the storage.

$$\phi_{H,sto} = \phi_{STO,th,0} + \phi_{H,gen} + \phi_{E,HP1} - \phi_{sto,l,nr} - [\phi_{H,DH} - \phi_{E,HP2}] \quad [Wh] \quad (75)$$

For charging and discharging processes, the tank is connected to two heat pumps. One heat pump upgrades the temperature at the return temperature of the tank if it is less than the current storage temperature at the bottom layer adding compression work to the tank during the process ( $\phi_{E,HP1}$ ). The second upgrades the temperature of the storage at the top layer when less than the current district heating supply temperature. This heat pump adds compression work ( $\phi_{E,HP1}$ ) to the district heating demand ( $\phi_{DH}$ ).

Auxiliary electricity consumed by the tank is related to that consumed by circulation pumps of heat pumps Eq. (76). The equations governing the hourly performance of heat pumps are presented in Section 2.3.1.

$$\phi_{E,sto,aux} = \phi_{E,HP1} + \phi_{E,HP2} \quad [Wh] \quad (76)$$

Thermal losses in the tank are calculated from the system of Eq. (77) [104]. In order to account for stratification phenomena, the tank is discretized in 10 layers of same thickness.

$$\phi_{sto,l,nr} = \sum_{k=1}^{10} K_{sto} \cdot A_k \cdot (T_{sto,k} - T_g) \quad [Wh] \quad (77)$$

In Eq. (77),  $k = \{1 \dots 10\}$  represents the number of stratification layers considered in the model.

The energy balance equation for every stratification layer and time step is presented in Eq. (78). In this equation the terms  $\phi_{sto,k,in}$ ,  $\phi_{sto,k,ext}$ , respectively, represent the inter and outer energy flows

between each one of these layers. The equation is solved with a mass flow of the top layer ( $k=10$ ) equal to that of the first heat pump supply line and a mass flow of the bottom layer ( $k=1$ ) equal to that of the heat pump return line.

$$\dot{m}Cp_k \cdot (T_{sto,k} - T_{sto,k,0}) = \sum \phi_{sto,k,in} - \sum \phi_{sto,k,ext} \quad [Wh] \quad (78)$$

The heat exchanges due to convection between layers is calculated in the following equation:

$$\phi_{sto,k \rightarrow k+1} = C_{1,sto,k \rightarrow k+1} \cdot A_{sto} \cdot (T_{sto,k} - T_{sto,k+1}) \quad [Wh] \quad (79)$$

where

$$C_{1,sto,k \rightarrow k+1} = \frac{Nu_{k \rightarrow k+1} \cdot k_w}{L_c};$$

$$Nu_{n \rightarrow n+1} = 0.27 \left[ \frac{g \cdot \alpha T_{sto} \cdot L_{car}^3 \cdot (T_{sto,k} - T_{sto,k+1}/2 - T_{sto,k}) \cdot \delta m_{sto}}{\nu_w^2 \cdot \delta T_{sto}} \right]$$

$$L_{car} = \frac{h_{sto}}{10}$$

The investment costs of the storage are presented in Eq. (80) accounting for the peak thermal load attended by the secondary heat pump in the supply line to the district ( $\phi_{HP2}^0$ ) and that attended by the main heat pump in the supply line to the storage ( $\phi_{HP2}^0$ )

$$InvC_{sto} = \left[ (-493.53 \ln C_{1,sto} + 5484) \cdot C_{1,sto} + 7229 \cdot V_{sto}^{0-0.522} \right] \cdot \varphi_{ann} \quad [EU] \quad (80)$$

where

$$V_{sto}^0 = \frac{\phi_{sto}^0}{Cp_w \cdot \rho_w \cdot (90 - 20)};$$

$$C_{1,sto} = \frac{(\phi_{H,gen,HP1}^0 + \phi_{H,gen,HP2}^0)}{1000}$$

O&M costs are derived from the resulting electrical needs of the heat pump and maintenance costs rated at 1% of total investment costs.

$$O\&MC_{sto} = 0.01 \cdot INVC_{sto} + \sum \frac{(\phi_{E,sto,aux}) \cdot \xi_{el,CH}}{1000} \quad [EU] \quad (81)$$

### 2.3.7. Substations and heat exchangers

Temperature and mass flow rate conditions of heat exchangers and substations were determined with the NTU method [105]. The NTU method expresses the effectiveness of heat transfer processes between a hot and cold stream in relation to their mass flow rate conditions, target temperatures of operation and type of heat exchanger unit. For all cases (i.e. solar collector heat exchanger, cooling and heating substations, server heat exchangers, refrigeration exchangers and sewage heat exchangers) we modeled a tube heat exchangers. In Eq. (82), the heat exchanger area is determined at nominal conditions of every customer substation. For this we assumed the nominal network temperature to be that maximum (or minimum in the cooling case) of all connected buildings at the peak hour plus a minimum delta of target temperatures of 5 K [4].

In the case of heat exchangers located in a sewage bed of length  $L_{HX,sw}$ , the area is determined from Eq. (83) accounting for the depth  $h_{HX,sw}$  of the active water in contact with the heat exchanger. Return temperatures and mass flow conditions are then derived from the NTU method in the system of equations (82)–(87). The

solution to this system of equations is outside of the scope of this article and can be found in [16].

$$\phi^0 = A_{HX} \cdot k_{HX} \cdot \Delta T_{ln}^0 \quad [\text{Wh}] \quad (82)$$

$$A_{HX} = L_{HX,sw} \cdot W_{HX,sw} \quad [\text{m}^2] \quad (83)$$

where

$$W_{HX,sw} = \arcsin \left( \frac{C_{1,HX}}{h_{HX,sw} + \left( C_{1,HX}^2 / 4h_{HX,sw} \right)} \right) \cdot \left( h_{HX,sw} + \frac{c^2}{4h_{HX,sw}} \right);$$

$$C_{1,HX} = \sqrt{\left( \left( R_{sw} - \frac{h_{HX,sw}}{2} \right) \cdot 8h_{HX,sw} \right)}$$

$$\varepsilon_{HX} = 1 - e^{1/C_r \text{NTU}^{0.22} \cdot e^{-C_r \text{NTU}^{0.78}}} \quad [-] \quad (84)$$

where

$$C_2 = \frac{C_{3,HX}}{C_{4,HX}}$$

$$C_{3,HX} = \min(\dot{m}_w c_{p_{WH}}, \dot{m}_w c_{p_{WC}})$$

$$C_{4,HX} = \max(\dot{m}_w c_{p_{WH}}, \dot{m}_w c_{p_{WC}})$$

$$\text{NTU} = \frac{A_{HX} \cdot k_{HX}}{C_{3,HX}} \quad [-] \quad (85)$$

$$\dot{m}_w c_{p_{WC}} = \frac{\varepsilon_{HX} \cdot (T_{H,in} - T_{C,in}) \cdot C_{3,HX}}{T_{C,out} - T_{C,in}} \quad [\text{kg/s}] \quad (86)$$

$$\dot{m}_w c_{p_{WH}} = \frac{\varepsilon_{HX} \cdot (T_{H,in} - T_{C,in}) \cdot C_{3,HX}}{T_{H,in} - T_{H,ext}} \quad [\text{kg/s}] \quad (87)$$

Based on the approach of [85] the investment cost associated to heat exchangers is derived as a function of the heat exchanger area in Eq. (88). O&M costs are presented in Eq. (89).

$$\text{INVC}_{HX} = 3364 \cdot \frac{A_{HX}^{0.6}}{5.5} \cdot \phi_{ann} \quad [\text{EU}] \quad (88)$$

$$\text{O\&M}_{HX} = 0.01 \cdot \text{INVC}_{HX} \quad [\text{EU}] \quad (89)$$

### 2.3.8. Circulation pumps

The hourly electricity consumption of circulation pumps is calculated according to Eq. (90). For this hourly mass flow rates and pressure losses ( $\Delta P_{Net}$ ) are estimated with the network optimization model of Section 2.4.1.

$$\phi_{E,pump} = \frac{\dot{m}_{net} \cdot \Delta P_{Net}}{\rho_w} \cdot \left( \frac{\dot{m}_{net}}{\dot{m}_{net}^0} \right)^3 \quad [\text{Wh}] \quad (90)$$

The investment costs of pumps are presented in Eq. (93). O&M costs are derived from the resulting electrical needs and maintenance costs rated at 1% of total investment costs in Eq. (92).

$$\text{INVC}_{pump} = [414.8 \cdot C_{1,pump}^{-0.187} + 440.11 \cdot C_{1,pump}^{-0.9,342}] \cdot \phi_{ann} \quad [\text{Wh}] \quad (91)$$

where

$$C_{1,pump} = \frac{\phi_{E,pump}^0}{1000}$$

$$\text{O\&M}_{pump} = 0.01 \cdot \text{INVC}_{pump} + \sum \frac{\phi_{E,pump} \cdot \xi_{el,CH}}{1000} \quad [\text{Wh}] \quad (92)$$

**Table 4**

Piping catalog of twin pipes for heating and cooling networks. \*  $r_i = 0.0018$ ; \*\*  $r_i = 0.0048$ .

| DN <sub>i</sub> [mm] | D <sub>i</sub> <sup>ext</sup> [mm] | D <sub>i</sub> <sup>int</sup> [mm] | Inv C <sub>p<sub>i</sub></sub> [EU/m yr] | $\dot{V}_{p,i}^{\min}$ [m <sup>3</sup> /s] | $\dot{V}_{p,i}^{\max}$ [m <sup>3</sup> /s] |
|----------------------|------------------------------------|------------------------------------|------------------------------------------|--------------------------------------------|--------------------------------------------|
| 20*                  | 26.9                               | 24.3                               | 1.2                                      | 0.0001                                     | 0.0005                                     |
| 25*                  | 33.7                               | 31.1                               | 1.3                                      | 0.0002                                     | 0.0008                                     |
| 32*                  | 43.4                               | 40.8                               | 1.5                                      | 0.0004                                     | 0.0033                                     |
| 40*                  | 48.4                               | 45.8                               | 2.1                                      | 0.0005                                     | 0.0041                                     |
| 50*                  | 60.3                               | 57.4                               | 2.8                                      | 0.0008                                     | 0.0065                                     |
| 65*                  | 76.1                               | 73.2                               | 4.2                                      | 0.0013                                     | 0.0105                                     |
| 80*                  | 88.9                               | 85.7                               | 6.2                                      | 0.0017                                     | 0.0202                                     |
| 100*                 | 114.3                              | 110.7                              | 9.5                                      | 0.0029                                     | 0.0337                                     |
| 125*                 | 139.7                              | 135.1                              | 14.7                                     | 0.0043                                     | 0.0502                                     |
| 150*                 | 168.3                              | 164.3                              | 20.8                                     | 0.0064                                     | 0.0742                                     |
| 200*                 | 219.1                              | 214.6                              | 25.0                                     | 0.0109                                     | 0.1266                                     |
| 250**                | 273                                | 268                                | 41.7                                     | 0.0169                                     | 0.1974                                     |
| 300**                | 323.9                              | 318.3                              | 50.0                                     | 0.0239                                     | 0.3183                                     |
| 350**                | 355.6                              | 350                                | 58.3                                     | 0.0289                                     | 0.3848                                     |
| 400**                | 406.4                              | 400.1                              | 66.7                                     | 0.0377                                     | 0.5029                                     |

## 2.4. Supply system optimization module

### 2.4.1. Thermal networks

Both layout and diameter of thermal networks are optimized to minimize investment and operation costs. Based on the work of [106,107], we have integrated a minimum spanning tree algorithm in the GIS platform ArcGIS to compute and represent a close-to-optimal layout for heating and cooling networks. For this, we selected a geometrical analysis grid consisting of roads, internal streets, pedestrian walkways and train tracks, which we connect orthogonally to a producer and consumer in the area. A main producer is selected between the building with largest storage area and the building with highest demand (Anchor load).

The pipe diameter of those close-to-optimal layouts is then determined so that the requested pressure at every customer is maintained between 119 kPa and 800 kPa. In a similar manner, the water velocity in every pipe is kept between acceptable limits (i.e. 0.3 m/s and the maximum volume flow per pipe type  $\dot{V}_{p,i}^{\max}$ ) and the volume flow rate, at every node and at every time span are met, while minimizing the total investment and operation costs. For this, we used the piping catalogue of Table 4 and assumed twin pipes and symmetrical pressure drop in supply and return lines. In this table, the rugosity coefficient includes 20% of losses due to fixtures. Investments costs are annualized at 60 years with an  $Ir=4\%$ , and include costs of both supply and return pipes.

We solved this hydraulic problem by integrating the non-linear convex formulations of [108] in the next mixed-integer linear problem (MILP):

$$O.F : 2 \cdot P_n^{\text{pl}} \cdot \dot{V}_n^{\text{pl}} \cdot \frac{\text{INVC}_{pump}}{(\dot{m}_{net}/\dot{m}_{net}^0)^3} + \sum_{e \in E} \sum_{i \in I} x_{i,e} L_{p_e} \text{INVC}_{p_i}$$

$$S.T : \dot{V}_{p,i,e} \geq x_{i,e} \dot{V}_i^{\min} \quad \forall i \in I, e \in E$$

$$\dot{V}_{p,i,e} \leq x_{i,e} \dot{V}_i^{\max} \quad \forall i \in I, e \in E$$

$$\sum_{i \in I} a_{k,i} \dot{V}_{p,i,e} + b_{k,i} \dot{V}_{p,i,e} \leq \frac{P_n - P_{n'}}{L_e} \quad \forall e \in E, k \in K$$

$$\sum_{e \in E^+} \sum_{i \in I} \dot{V}_{i,e} - \sum_{e \in E^-} \sum_{i \in I} \dot{V}_{p,i,e} + \dot{V}_n^{\text{rq}} = 0 \quad \forall n \in N \setminus n^{\text{pl}}$$

$$\sum_{i \in I} \dot{V}_{p,i,e} = 1 \quad \forall e \in E$$

where

$$x_{i,e} \in Z \quad \forall i \in I, e \in E$$

$$0 \leq x_{i,e} \leq 1 \quad \forall i \in I, e \in E$$

$$P_n^{\min} \leq P_n \leq P_n^{\max} \quad \forall n \in N$$

Following the recommendations of the author, we solve the MILP with a linearization-based branch and cut algorithm. In this algorithm, the function of pressure drop of Darcy–Weissbach  $h_{f_i}(\dot{V})$  in Eq. (93) is linearized in order to avoid a singularity in the friction coefficient  $f_D$ , of Swamee–Jain in Eq. (94). For this linearization process we assume that  $h_{f_i}(\dot{V})$  is non-linear convex under the condition  $\dot{V}_i^{\min} \leq \dot{V}_{p,i,e} \leq \dot{V}_i^{\max}$  and approximate  $h_{f_i}(\dot{V})$  in Eq. (95) with the linear cuts  $a_{k,i}$  and  $b_{k,i}$  described in Eqs. (96) and (97).

$$\frac{\Delta P_{\text{Net}}}{L} = h_{f_i}(\dot{V}) = \frac{8f_D \rho_w}{\pi^2 D_i^3} \dot{V}^2 \quad [\text{Pa/m}] \quad (93)$$

$$f_D = 0.25 \left[ \log_{10} \left( \frac{R_i}{3.7D_i} + 5.74Re^{-0.9} \right) \right]^{-2} \quad [\text{m}] \quad (94)$$

where

$$Re = \frac{4\dot{V}}{\pi v_w d_i}$$

$$\dot{V}_{k,i}^{\text{linear}} = \dot{V}_i^{\min} + \frac{\dot{V}_i^{\max} - \dot{V}_i^{\min}}{K - 1} (k - 1) \quad [\text{m}^3/\text{s}] \quad (95)$$

$$a_{k,i} = \frac{dh_{f_i}}{d\dot{V}}(\dot{V} = \dot{V}_{k,i}^{\text{linear}}) \quad [-] \quad (96)$$

$$a_{k,i} \dot{V}_{k,i}^{\text{linear}} + b_{k,i} = h_{f_i}(\dot{V}_{k,i}^{\text{linear}}) \quad [-] \quad (97)$$

In a first step the pressure losses for the optimized layout are plotted against mass flow rates of the network and the result is linearized in Eq. (98). Similarly, the investment costs of the optimal layout are linearized in proportion to the number of buildings to be connected in Eq. (99). Both equations serve as main input for the optimization routine of Section 2.4.2.

$$\Delta P_{\text{net}}^{\text{linear}} = 2 \cdot (C_{1,\Delta P} \cdot \dot{m}_{\text{DH/DC}} + C_{2,\Delta P}) \quad [\text{Pa}] \quad (98)$$

$$Inv_{\text{net}}^{\text{linear}} = Inv_{\text{DH/DC}} \cdot \frac{\sum_{B \in B^c} b}{\sum_{B \in B} b} \quad [\text{EU}] \quad (99)$$

Heat losses in the network ( $\phi_{\text{dis},l,\text{nr}}$ ) are estimated according to the following equation [85]:

$$\phi_{\text{dis},l,\text{nr}} = (T_{\text{DH/DC},\text{sup}} - T_x) * \overline{m_{\text{wcp}_w}} \quad [\text{Wh}] \quad (100)$$

where

$$T_x = T_g + (T_{\text{DH/DC},\text{sup}} - T_g) * e^{-K_p * L_p / \overline{m_{\text{wcp}_w}}}$$

where  $\overline{m_{\text{wcp}_w}}$  is the average value between the maximum and minimum flow running in the network. The temperature of supply required by the central plant  $T_{\text{DH/DC},\text{sup}}$  is then recalculated to account for losses.

#### 2.4.2. Size and operation optimization

Based on the approach of [48,109], a bi-level optimization strategy was used to configure a community energy system for minimum total annual costs (TAC), emissions ( $\text{CO}_2$ ), and primary energy needs (PEN). The problem is subjected to constraints of resource availability (Section 2.2), systems technology operation (Section 2.3) and network configuration (Section 2.4.1).

The multi-objective and multi-period (hourly) optimization problem is formulated as a mixed-integer non-linear program and is divided into a Master and a Slave routine where:

$$O.F : \min_{x \in E} f(x) = \text{TAC}(x), \quad \text{GHGE}(x), \quad \text{PEN}(x)$$

$$S.T : \text{TAC}(x), \text{GHGE}(x), \text{PEN}(x) = \text{Slave}(x)$$

where

$$\text{TAC}(x) = \text{INV}(x) + \text{O\&M}(x)$$

The master routine consists in an evolutionary algorithm, which defines a system set-up and mutates it in regards to the previously found best alternative. The system set-up consists of a network layout with a set of connected customers, and combination of different production and storage units. The supply system of disconnected customers is optimized with both Master and Slave routines in a previous time step. For this last case, the optimization problem does not consider network set-up, but small-scale technologies such as fuel cells, natural gas-fired boilers, biogas-fired boilers, GSHP and solar technologies.

The slave routine evaluates the operation of each system set-up upon the three-optimization variables of concern (i.e. TAC,  $\text{CO}_2$  and PEN). In line with the work of [110] we chose a fixed order operation model for activation of production units. In this model, technologies connected to endogenous resources are offered at zero marginal cost, whereas other production units are forced to adapt their operation scheme on demand. Surplus energy from a zero-cost source is then stored with an efficiency penalty (i.e. simulated storage thermal loss). Solar technologies are prioritized followed by sources of waste heat. Then heat pumps are activated according to its hourly efficiency followed by co-generation units, condensing boilers and peak boilers. As a rule, cogeneration units operate at a minimum of 2000 h and not less than the part load factors of Tables 2 and 3. The last constrain applies for all technologies.

Once the results of installed capacities of solar technologies and grid connectivity are known for each Pareto-optimal configurations, the model of Section 2.4.1 is reapplied to those configurations.

#### 2.5. Decision module

##### 2.5.1. Reliability assessment

Following the approach of [48,111], we conducted a sensitivity analysis to characterize the reliability of Pareto-optimal configurations and determine influential parameters of the model. For this, we evaluated changes in the price of natural gas, biogas and electricity in the bandwidth of  $\pm 10\%$  with increases of 2%. For every increase, the Pareto-front is re-evaluated, and the Pareto-dominant configurations are selected. The reliability level is measured as the relative number of times a configuration is Pareto-dominant in relation to other configurations. The parameters with the highest associated number of discarded configurations are the most sensible.

##### 2.5.2. Multi-criteria decision analysis

In order to select configurations that either maximize the social, economic and environmental value of the are, we integrated the multi criteria decision analysis (MCDA) approach of [112]. For this we considered three criteria levels (i.e. social, economic, and environmental) with weight  $w_i$ , and six sub-criteria (i.e. social: penetration of renewables, export of electricity, economic: total annual energy costs, annual operational costs, environmental: emissions



and non-renewable primary energy) with weight  $w_{i,j}$  subjected to the conditions of the following equation:

$$\begin{aligned} \sum_{i=1}^3 w_i &= 1, \quad \forall i \in \{1, 2, 3\} \\ \sum_{j=1}^2 w_{ij} &= 1, \quad \forall j \in \{1, 2\} \end{aligned} \quad (101)$$

Each Pareto-optimal configuration  $k$ ;  $k \in \{A, B, C\}$  is rated for all sub-criteria. The rating values are labeled  $X_{ij}^k$ . For each sub-criterion, the best and the worst ratings are determined in Eqs. (103)–(104) and normalized in Eq. (105).

$$\begin{aligned} \sum_{i=1}^3 w_i &= 1, \quad \forall i \in \{1, 2, 3\}; \\ \sum_{j=1}^2 w_{ij} &= 1, \quad \forall j \in \{1, 2\} \end{aligned} \quad (102)$$

$$X_{ij}^{\text{best}} = \min_{K \in A, B, C} X_{ij}^K, \quad \forall i \in \{1, 2, 3\}, \quad \forall j \in \{1, 2\} \quad (103)$$

$$X_{ij}^{\text{worst}} = \max_{K \in A, B, C} X_{ij}^K, \quad \forall i \in \{1, 2, 3\}, \quad \forall j \in \{1, 2\} \quad (104)$$

$$\bar{X}_{ij}^K = \frac{X_{ij}^K - X_{ij}^{\text{worst}}}{X_{ij}^{\text{best}} - X_{ij}^{\text{worst}}} \quad (105)$$

The final score per configuration is obtained in Eq. (106). The best option is that with a value of one (1).

$$\text{Score}_k = \sum_{i=1}^3 \sum_{j=1}^2 w_i \cdot w_{ij} \cdot \bar{X}_{ij}^k \quad (106)$$

### 2.5.3. Life cycle analysis

We integrated the method of the energy path of Switzerland for life cycle analysis of buildings [113]. For this, all energy vectors related to construction/dismantling, operation, and induced mobility are evaluated. Those vectors related to the operation of buildings are calculated with the procedure presented from Sections 2.1–2.4. Emissions and primary energy related to construction or dismantling of buildings (embodied energy and grey emissions) are calculated according to the approach of [5]. The method consists in a relation of the database of grey energy and emissions of [114] to the occupancy type, construction year, retrofit year and built area of buildings. These data is extracted from the urban database presented in Section 2. Finally, vectors related to mobility are estimated based on the recommendations of [113]. Similar to the approach of [5], a database of transportation means and factors of emissions and primary energy consumption is correlated to the occupancy type and built area of buildings.

### 2.5.4. Benchmarking

We benchmark the performance of buildings by comparing the results of Pareto-optimal configuration to local averages of end-use energy consumption, and target values of carbon footprint and primary energy consumption. Demand side benchmarks are derived from the database of archetypes of [1]. This database contains data related to specific use of electricity, heating and cooling services in buildings. The database is correlated to the occupancy type and built area of buildings to determine a comparable value. Carbon foot-print and primary energy benchmarks are extracted from [115] and contain benchmarks for embodied energy, operation of buildings and induced mobility.

## 2.6. Spatiotemporal analysis module

We expanded the approach of [1] and developed a 4D interface to represent time-dependent flows of energy between buildings and infrastructure. The approach was demonstrated in [1] to serve as a catalyst of discussions between decision makers. In this project, this visual interface serves in addition as a feedback loop (see Fig. 1) for energy systems designers and urban planners, who by mapping final infrastructure solutions, patterns of demand and resource availability could start making the modifications necessary to translate this configurations into more market-adaptable solutions. For instance, by improving manually the expected layout of energy networks or changing the efficiency of infrastructure components. Other modifications could lie on urban planning decisions targeting to evaluate impacts and benefits to potential future users. For instance discussing if parcel of  $x$  end-user optimized for photovoltaic technology does match the financial or even personal needs of its owner. We present an example of this technique in Figs. 5 and 6.

## 3. Test case

We tested the CEA framework in a real case study in the Swiss city of Zug. It consists of an industrial site of 25 ha undergoing a process of urban transformation. A large manufacturer in the industry sector owns and predominately occupies the site along with other companies in the services sector. At the moment, there is no residential use on site. Departing from today's condition or Status Quo scenario (SQ), a past research project [5] developed four retrofit scenarios for the area by 2035. The Business-as-Usual scenario (BAU) focuses on the industrial legacy of the site with a small expansion to residential uses. The Urban Condenser scenario (UC) describes a classic city-district development with a mix of industrial, residential and commercial uses. It consists in a more inclusive and socially driven strategy to increase the liveability of the area, offer housing close to job opportunities and open spaces for leisure and recreational activities. The High-end Business scenario (HEB) drives the light industry out of the district and populates it with high-rise buildings with commercial use. In this scenario, priority is given to the tertiary sector of the economy, which by many would guarantee the economic sustainability of urban areas in the future. On the other hand, the Campus scenario (CAMP) describes a university environment with laboratories, housing and catering services. In this area, students, residents and workers meet in close vicinity of big parcels of open space, which evoke for total pedestrian friendly mobility.

### 3.1. Demand assessment

We used the demand module of Section 2.1 to calculate the future demand of end-uses in buildings. Ten types of end-uses<sup>3</sup> were estimated.

According to Fig. 3a, the heating demand in the area could potentially increase as a result of higher use of domestic hot water in new residential areas. The cooling demand could potentially increase despite needs of industrial processes and server rooms disappear. Electrical loads would slightly increase in scenarios with high office space and industrial production (i.e. SQ, BAU and HEB), whereas it could decrease in cases with more residential and mixed uses (i.e. UC, CAMP).

<sup>3</sup> Qcp: process cooling, Qhp: process heat, Qhs: space heating, Qcs: space cooling, Eal: appliances & lighting, Ep: industrial processes (electricity), Edat: server room (electricity), Ecair: compressed air (electricity), Qww: domestic hot water, Qcdata: server room cooling.

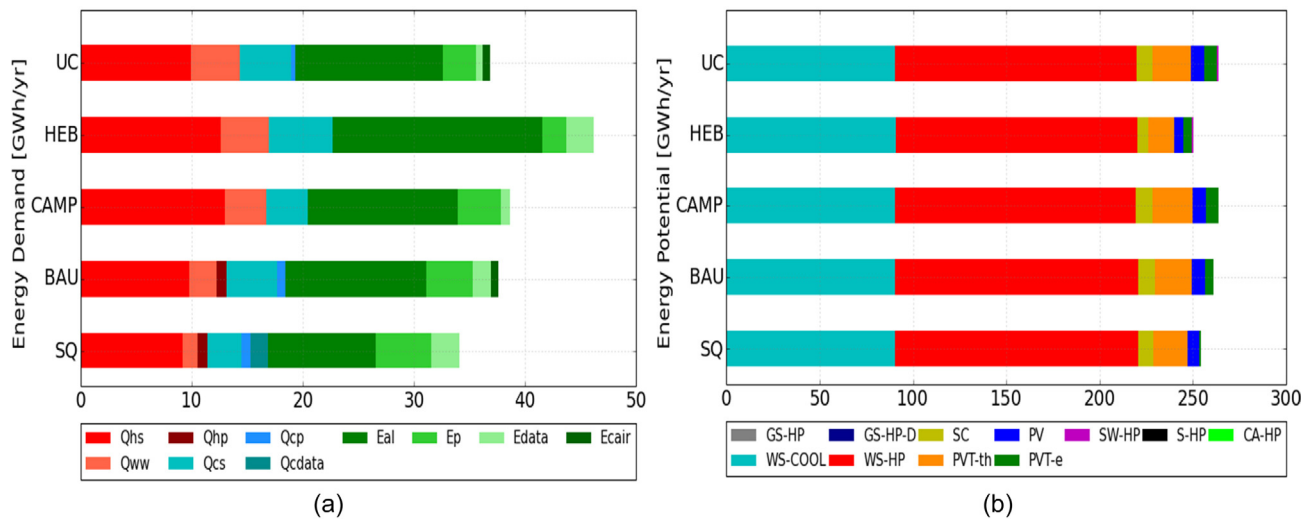


Fig. 3. (a) yearly energy demand per scenario, (b) yearly energy potential per scenario.

In Fig. 12 of Annex A, we estimated the hourly demand of heating, cooling and electrical services for every future scenario. The peak demand of heating, cooling and electrical services increases in all scenarios and trends to be doubled: the first, increases in the range of 10–100% (11–20 MW); the second, increases in the range of 25–122% (6–10 MW); the third increases in the range of 13–100% (4.5–8 MW). The HEB scenario presents the highest peak of all scenarios for all energy services. This behavior is attributed to high volumes of office and commercial space in combination with high window/wall ratios; air-based conditioning systems and data related cooling loads. For cooling and heating services, the lowest peaks take place in the UC scenario. Despite presenting a similar yearly cooling demand to the BAU scenario, and higher in the heating case, the high mix of air-conditioned uses (offices, industry & commercial) particular of the UC scenario generates close to 15% less peak heating and cooling demand (Fig. 12). For electrical services, a similar behavior is observed. In this case the UC scenario presents a peak 20% inferior to scenarios with similar yearly electricity demand (i.e. BAU and CAMP scenarios). However, this peak is presented up to three times in a day as a consequence of the diversity of uses previously mentioned of this scenario. The mid-day peak during summer (Fig. 12) is particularly interesting for the integration of solar technology.

### 3.2. Resource availability assessment

From the analysis of the demand we observed a potential use of waste heat from industrial processes (compressed air), sewage, and data centers. We used the module of Section 2.2 to calculate the availability potential of these and other endogenous resources per scenario.

For the lake-water potential, we considered an existing intake on site at 33 m below ground and obtained temperatures at the source oscillating between 4 and 6 °C around the year. In this order of ideas the lake represents one of the best sources for cooling services in the area. However, for environmental reasons, the use of lake as a source of heat is constrained so that at the end of the year its energy content does not change above 12,500 MW h. The maximum extraction rate from the existing installation in site could have a maximum intake of 2000 m<sup>3</sup> [116].

The use of geothermal energy in the area is totally restricted with the exception of new tall-buildings (more than 20 floors) with energy piling technology. Based on observations in the area we assumed new tall-buildings to have an average piling depth

of 40 m (i.e. maximum depth of exploration), an average piling distance of 5 m<sup>4</sup> (average separation between probes) and a maximum geothermal capacity per pile of 50 W<sub>th</sub>/m. These restrictions are integrated in the model to determine the maximum potential of geothermal energy per scenario.

On the other hand, we calculated the solar potential of roof top areas and estimated the energy yield per solar technology. We used the module of Section 2.6 to represent changes in solar insolation and obtain a description of the solar potential in every building and scenario. In Fig. 5 the results of the HEB scenario are presented.

We estimated the availability of waste heat from datacenters, compressors and sewage lines with the models of Section 2.2.3. Finally, the potential of fuel-fired technologies (i.e. condensing boilers, combined gas cycle turbines and fuel cells) is estimated in relation to the maximum availability of fuel in the local grid (i.e. 15 MW<sub>th</sub>) with the models of Sections 2.3.2 and 2.3.3.

In Fig. 3b we present the share of local energy potentials available per scenario and in Fig. 13 of Annex A, we present their time variability. The potential of lake-water heat for either heating or cooling is one of the highest and more stable for the area. The hourly power availability of this source at a typical network temperature of 70 °C for heating purposes and 8 °C for cooling purposes lies, respectively, in 15 MW and 10 MW during the whole year. This relatively stable source/sink of heat could supply the total needs of most scenarios (with the exception of HEB for heating purposes). In the heating case, the lake-water presents a peak power capacity highly comparable to the current availability of natural gas from the local utility. Despite the high temperature lift and low COP expected while using this source for heating purposes (i.e. 3.2–3.7), its carbon emissions and primary energy consumption could be, respectively, 40% and 85% less than that of a Natural-gas fired boiler.

After lake-water, solar technologies present the highest potential on site. In terms of thermal yield, this potential oscillates between 4 GW h/yr (the lowest presented with solar collectors for HEB scenario) to 20 GW h/yr (the highest presented with photovoltaic/thermal for CAMP scenario). This relatively big difference lies on one hand on the low availability of roof areas of HEB scenario with respect to the CAMP scenario. On the other hand, it lies on the high performance of PV/T technology, whose low

<sup>4</sup> This value represents a high vicinity between boreholes, which represents a 10% less yearly efficiency in return for more capacity of installation [127].

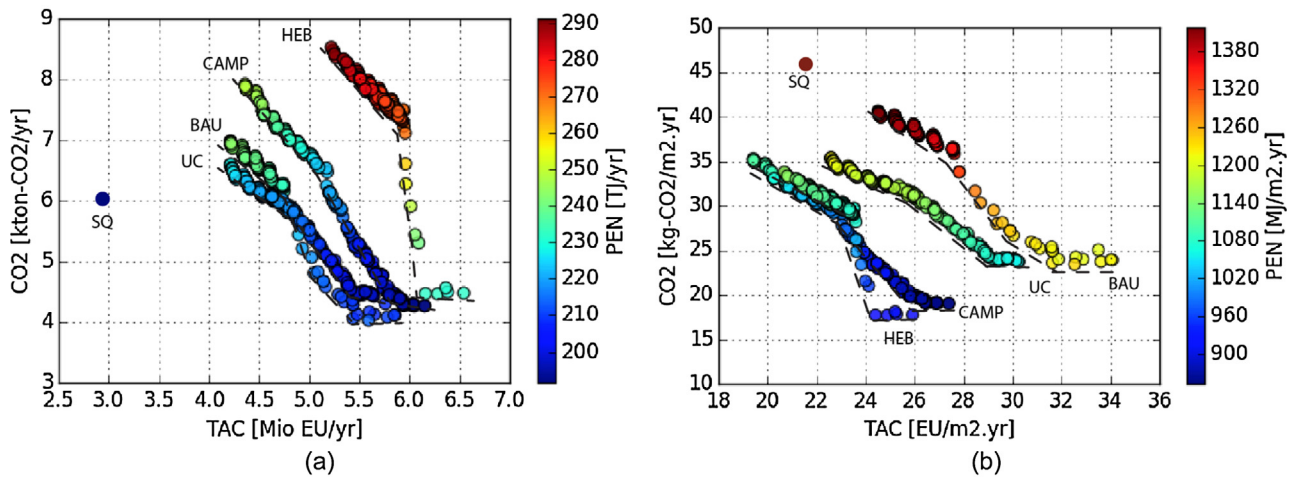


Fig. 4. Comparison of Pareto-frontiers after 24 generations, (a) total values, (b), relative values.

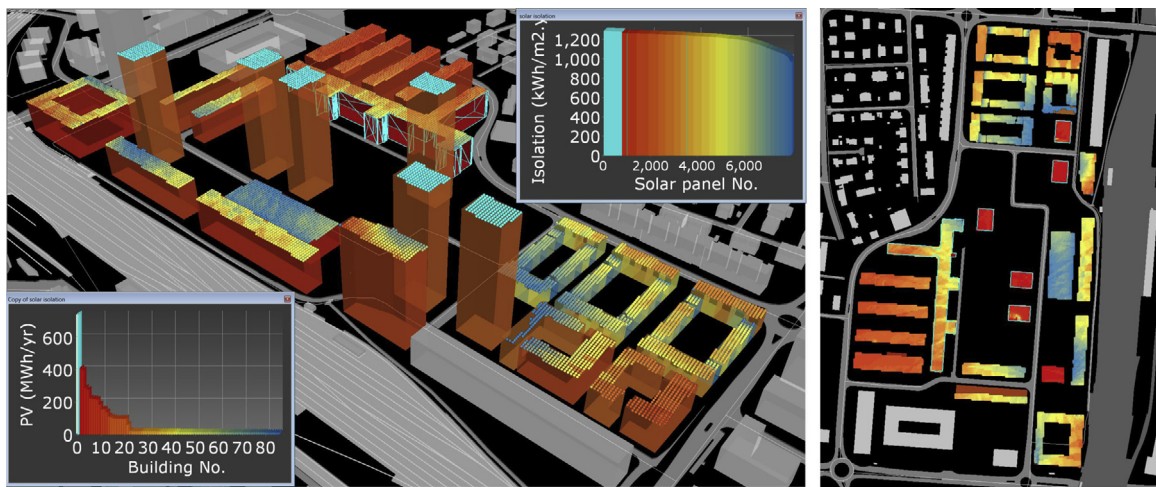


Fig. 5. Yearly solar isolation HEB scenario.

operating temperature translates into significantly higher thermal energy yield.

Another interesting potential energy source especially for BAU and CAMP scenarios is waste heat from sewage. This low-quality source of energy has the capability to cover close to 10% of the peak heating demand of both scenarios and cover close to 9 GW h/yr (from 5% to 10% of the yearly heating demand on site). Server rooms (1.4 MW h/yr, 200 kW) and process heat (0.60 MW h/yr, 63 kW) represent the lowest potential together with geothermal. For both decentralized and centralized options geothermal could cover, respectively, 30 kW and 45 kW of peak heat energy.

In terms of electricity generation, PV is the most suitable alternative besides cogeneration. It achieves an average yield between 4 GW h/yr and 7 GW h/yr, whereas centralized cogeneration units (combined cycle) have the capacity to generate up to 61 GW h of useful electricity). The question is: how much of these potentials can be feasibly integrated?

### 3.3. System's optimization

To answer this question the optimization routine of Section 2.4 is used to determine Pareto-optimal configurations per scenario. In Fig. 6 presents one of these configurations for the HEB scenario where all potential customers are connected through a thermal

micro-grid. Peak demands, mass flow rates and network layout are mapped to furthermore investigate potential “bottlenecks” and/or ill-conditioned nodes (those with a relative high mass flow rate). The last could be manually changed prior a new iteration.

We applied the reliability assessment method of Section 2.5.1 to later extract the most robust solutions in Fig. 4. For this we used a reliability threshold of 95%. As a general consensus, the algorithm discards close to 10% of all individuals of the Pareto-front. The most sensible variable of the analysis consisted in the price of natural gas. This result constrains the feasibility of combined heat and power units and natural gas-fired boilers when selected.

According to Fig. 4a, all future trajectories of development will require an increase in the total annual costs of at least 45%. Besides a few alternatives in the CAMP and HEB scenarios, the total annual cost per unit of area (Fig. 4b) could increase up to 60%. These values are more related to the type of urban trajectory rather than to the system configuration. In terms of relative values, both emissions and primary energy consumption might decrease in the area (Fig. 4b). In the worst case (BAU scenario) emissions could decrease between –5% and –46%, whereas in the best, it could decrease between –12% and –61% (HEB scenario).

In terms of global values, emissions could either decrease up to –27% (UC scenario) or increase in 35% (HEB scenario) for the area. Most environmentally performing alternatives require financial



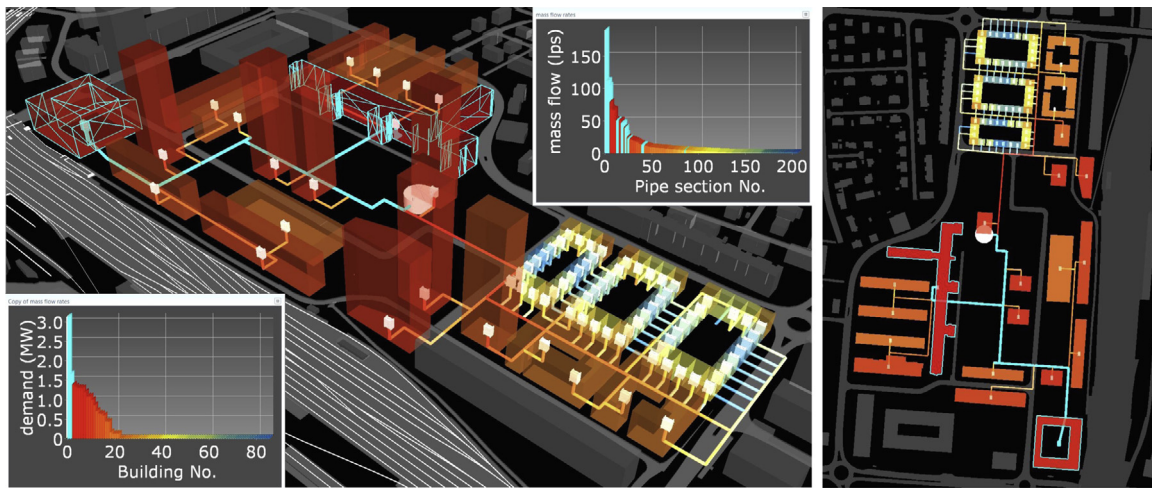


Fig. 6. Infrastructure and performance patterns for selected configuration of scenario HEB. 12:00pm December 12th.

means in the range of 26 EU/m<sup>2</sup> yr. If we consider the average area per new household in Switzerland (i.e. 100 m<sup>2</sup>)<sup>5</sup>, this number would translate in an average burden of 215 EU/household/Month per heating, cooling and electricity services. This number goes in line with current living costs charged to renters in the Zug area (i.e. 200–300 EU/household/Month). This point of view makes even the most expensive alternative plausible. A discussion of a proper business model in this direction is outside of the scope of this article but is presented as a potential alternative to the integration of highly carbon-efficient infrastructure configurations.

### 3.4. Multi-criteria analysis

We conducted a multi-criteria analysis (see Section 2.5.2) in an intra-scenario and inter-scenario basis to select the most suitable Pareto-optimal configuration and scenario for the area. We assumed this condition as the one with equally distributed weights among all sub-criteria (i.e. societal: %RES, economical: LEC, TAC, environmental: CO<sub>2</sub>, PEN).

An alternative to this procedure would consist on selecting strategies based on maximum available budget or expenditure willingness. Whereas a maximum available budget consists in selecting alternatives below a threshold of maximum investment costs, the expenditure willingness can be portrayed with relative metrics of TAC.

In order to map preferences for societal, economic or environmental values, we selected a series of Pareto-optimal configurations per scenario where each value is weighted at 80% over the others at the time. We then compared the estimated savings for each sub-criteria with respect to the existing configuration in the area (district heating network assisted by central Natural gas-fired boiler, Lake-water cooled district cooling network and direct connection to regional electricity grid). As presented in Fig. 7 and for a balanced distribution of all criteria, the results showed potential savings from 40% to 50% in emissions and from 25 to 30% in primary energy at an annualized cost between 18% and 35% higher than today. For an economic-driven distribution, the results showed savings of up to 4% in emissions, 14% in primary energy and 5% in costs. For an environmentally- so socially-driven distributions, the estimated savings in emissions and primary energy are comparable to those of the balanced distribution.

In terms of an inter-scenario comparison, we took the series of Pareto-optimal configurations previously selected and compared them to the current performance of the SQ scenario. As presented in Fig. 8 and for a balanced distribution of all criteria, the results showed relative savings from 45% to 60% in emissions and from 25 to 50% in primary energy at an annualized cost between 13% and 44% higher than today. For an economic-driven distribution, the results showed savings of up to 23% in emissions, 36% in primary energy and 11% in costs. For an environmentally- so socially-driven distributions, the estimated savings in emissions and primary energy are comparable to those of the balanced distribution.

From the last analyses two conclusions can be withdrawn: the first is that total emissions, primary energy consumption and related operational costs are due to increase in the area. In relative terms, those quantities could be lower in regards to the current status quo what translates into a potential increase in energy performance.

A second conclusion concerns the form and morphology of every urban trajectory under analysis. From a comparison of Figs. 7 and 8 we identified that relative savings up to 15% in emissions and 20% in primary energy with variable costs between up to 23% in the area are related to the urban design option rather than to its energy systems configuration. Further studies in this direction should be carried out in order to generalize such conclusion.

On the other hand, in Tables 5 and 6 we present performance indicators and technical characteristics of the balanced configuration in an scenario basis. These configurations prioritize the use of systems at minimum marginal cost (i.e. lake-water pumping, PV and PVT systems) and of biogas over natural gas (for boilers). Fuel cells, solar collectors and combined cycle equipment appeared too expensive to be competitive in the optimization process. In addition, with more than 50% of the buildings connected to the district-heating network in average, the optimization favors a combined (totally and partially distributed) operation for the energy supply.

#### 3.4.1. Life cycle analysis and benchmarking

We evaluated the full environmental performance of the case study in line with the targets of the 2000-W society vision in Switzerland. For this, we compared the emissions and primary energy consumption of the chosen Pareto-optimal configuration due to buildings and infrastructure to those due to production processes, logistics, and commuting, business flights and embedded energy in buildings. The last data is gathered from a previous

<sup>5</sup> Assuming a density of 2.5 persons per household [128], and 40 m<sup>2</sup> of built area per person (extracted from demand model of Section 2.1).

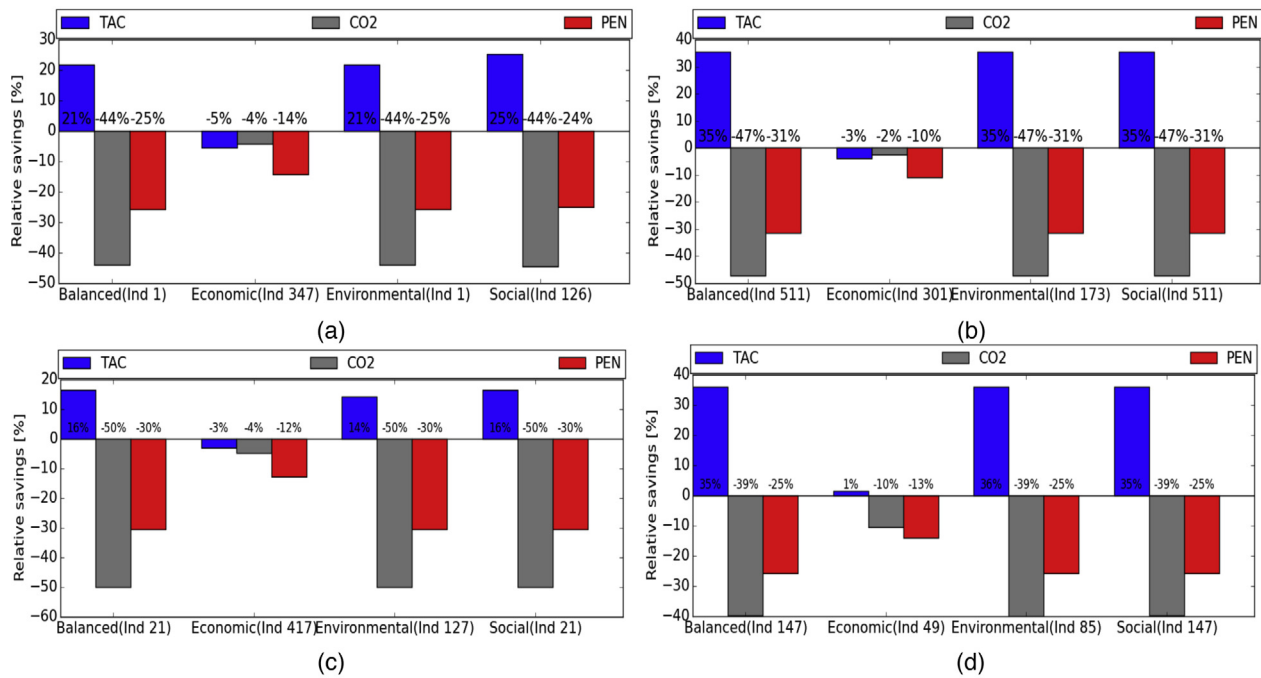


Fig. 7. Intra-scenario multi-criteria assessment for balanced, economic, environmental and social driven criteria, (a) BAU, (b) CAMP, (c) HEB, (d) UC.

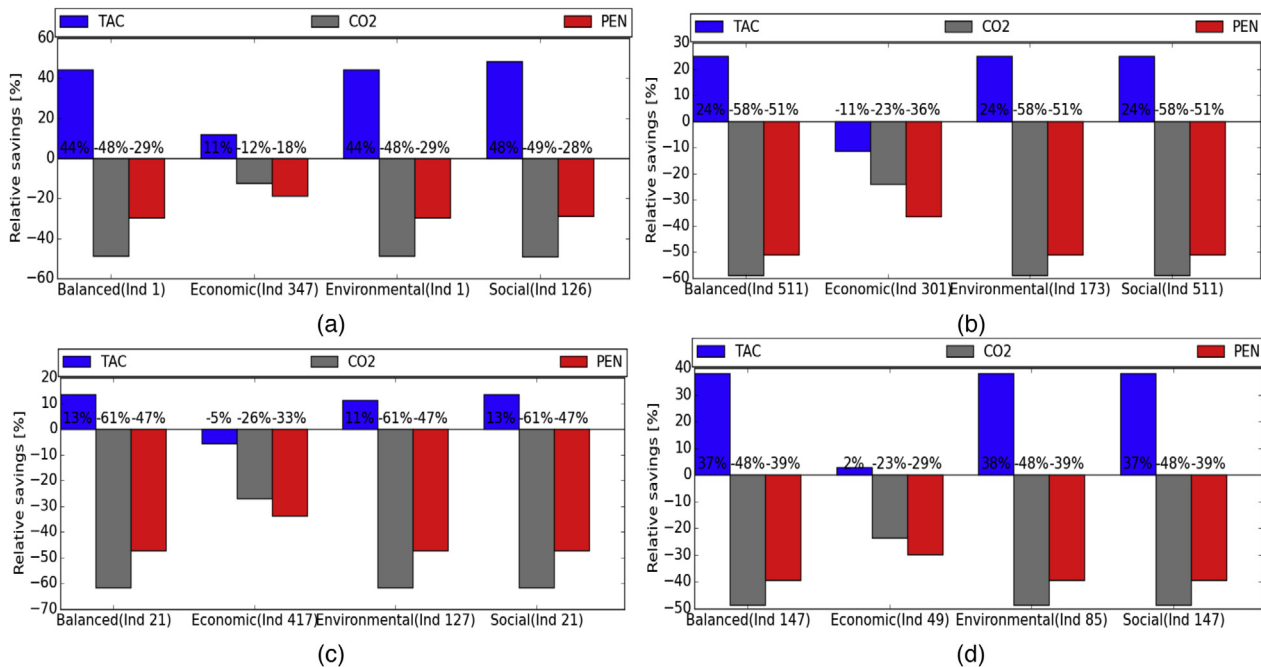


Fig. 8. Inter-scenario multi-criteria assessment for balanced, economic, environmental and social driven criteria, (a) BAU, (b) CAMP, (c) HEB, (d) UC.

research project in the area of study [5]. Based on the approach of [117] we gathered target values of the 2000-W-society vision for the year of 2050. These target values are obtained for the dimensions of embodied energy, operation of buildings and mobility. As shown in Fig. 10, both, grey emissions and embodied energy in buildings increases for every scenario at a maximum of 18% from today's levels, still below the target values. For the Swiss average, we might assume that an urban transformation in this direction would most likely increase its carbon footprint due to embodied energy.

For operation of buildings and industry, we found the expected performance of each scenario (Fig. 9b) for the selected

Pareto-optimal configuration to slightly exceed its target values due to primary energy consumption while still being within the limits of CO<sub>2</sub> requirements (Fig. 9d). This behavior is strongly related to factors such as technology efficiency and the Pareto-optimal configuration selected in Section 3.1.4 rather than to industry operation, whose share is negligible in comparison to building operation (Fig. 9). As in the total values of Fig. 10, the predominant share of emissions belongs in business flights whereas the lowest lie in logistics and industrial processes. The total emissions in the area can decrease up to 48% for a scenario with low business flights, but increase in 32% for the most demanding one. In terms of emissions due to mobility the panorama can highly change



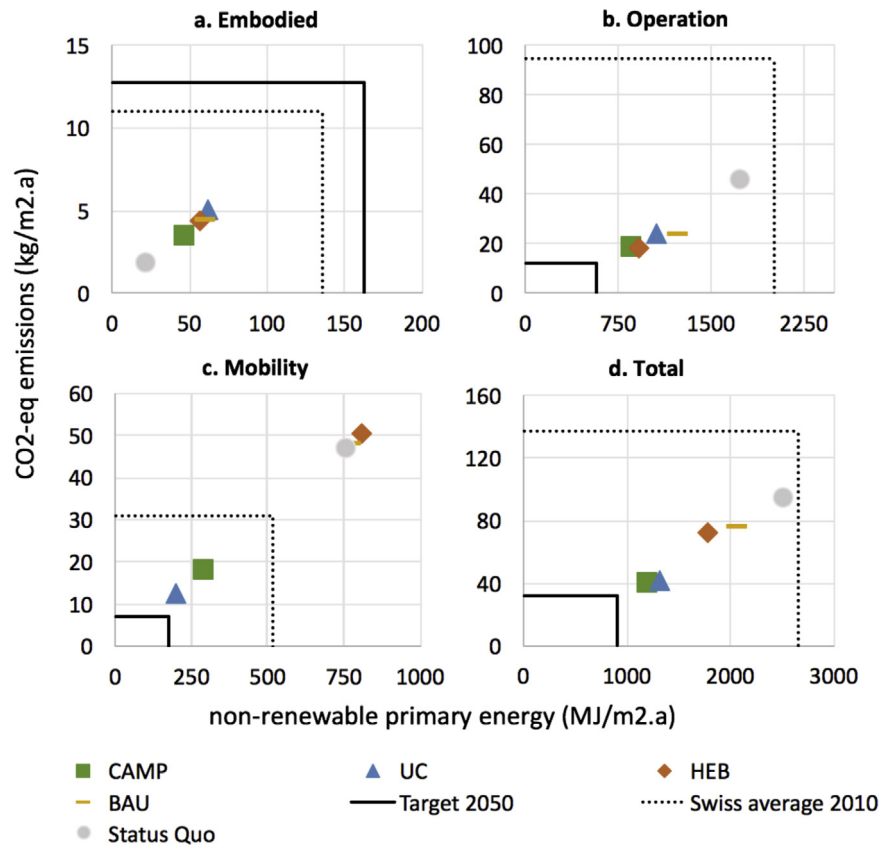


Fig. 9. Comparison to medium term 2000-W/1-t CO<sub>2</sub> society goals (2050) due to: (a) embodied energy, (b) building operation, (c) mobility and (d) total for the area.

**Table 5**  
Performance indicators for best configuration.

|                                                            | BAU  | CAMP | HEB  | UC   |
|------------------------------------------------------------|------|------|------|------|
| %RES [%]                                                   | 52   | 39   | 46   | 38   |
| CO <sub>2</sub> [kg CO <sub>2</sub> -eq/a/m <sup>2</sup> ] | 24   | 19   | 18   | 24   |
| PEN [MJ-oil-eq/a/m <sup>2</sup> ]                          | 1223 | 853  | 921  | 1056 |
| LEC [CHF/kWh]                                              | 0.14 | 0.16 | 0.13 | 0.15 |
| TAC [CHF/a/m <sup>2</sup> ]                                | 32   | 27   | 25   | 30   |

**Table 6**  
Characteristics for the best configuration. \* Existing infrastructure on site.

|                                                  | BAU   | CAMP  | HEB   | UC    |
|--------------------------------------------------|-------|-------|-------|-------|
| Base/peak boilers NG [MW]                        | –     | 10.2* | –     | 7.6*  |
| Individual boilers NG [MW]                       | 0.49  | 2.5   | 2.9   | 3.9   |
| HP Lake [MW th]                                  | –     | –     | –     | –     |
| HP sewage [MW th]                                | –     | –     | –     | –     |
| Heat recovery industrial processes [MW]          | 0.2   | 0.2   | 0.2   | 0.2   |
| Pump–Lake [MW]                                   | 0.36* | 0.24* | 0.45* | 0.33* |
| HP ground [MW]                                   | –     | –     | –     | –     |
| SC [MW th]                                       | –     | –     | –     | –     |
| PV [MW el, peak]                                 | 0.38  | –     | –     | –     |
| PV-T [MW el, peak]                               | 3.68  | 2.8   | 3.29  | 2.2   |
| PV-T [MW th]                                     | 14.6  | 11.1  | 14.2  | 8.4   |
| Storage [× 10 <sup>3</sup> m <sup>3</sup> ]      | 89.1  | 73.4  | 92.3  | 65.6  |
| HP storage [MW th]                               | 6.8   | 5.2   | 6.2   | 6.5   |
| Roof-top solar area (of connected buildings) [%] | 67    | 48    | 96    | 47    |
| Roof-top solar area [%]                          | 65    | 42    | 81    | 34    |
| Buildings connected [%]                          | 82    | 93    | 56    | 62    |

from scenario to scenario. In general SQ, BAU and HEB scenarios could potentially generate up to four times more emissions due to mobility (Fig. 9c). This behavior is driven by top-level management positions and related business flights, which correspond to

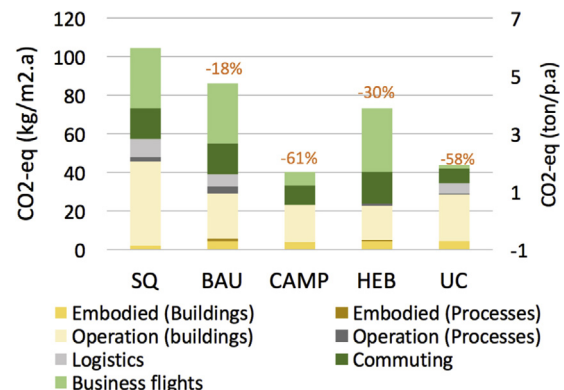


Fig. 10. Variation in percentage of CO<sub>2</sub>-eq emissions.

the highest share of total emissions in the area (Fig. 10). In consequence, the HEB scenario will be the most polluting one with 32% more emissions per unit of area than today (SQ).

As shown in Fig. 10 the CO<sub>2</sub>-eq emissions per scenario will decrease by 20% to 63%, the later coming close to the 2000-W/1-ton CO<sub>2</sub> benchmark of 75% (2 t/p.a). The highest leverage to attain this benchmark for the area of concern exists in the building sector, where despite the grey emissions of new buildings and infrastructure, technical advances in conversion technologies would allow to drastically reduce emissions. This is the case of an increase from today's 16% efficiency of photovoltaic technology to 20%, which would allow the CAMP and UC scenarios to attain the target value of total emissions for the area. For HEB and BAU scenarios, these target values are quite difficult to attend as they are mostly influenced by their high demand of flight mobility.

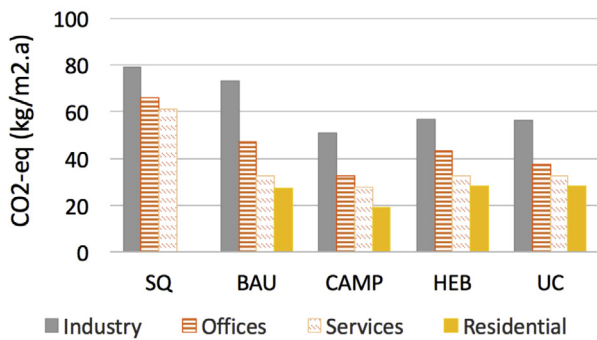


Fig. 11. CO<sub>2</sub>-eq emissions per building sector.

As depicted by the relative values of Fig. 11, the highest CO<sub>2</sub>-eq savings are found in residential, offices and services sectors while the lowest in industry. This behavior is due to the relatively few energy efficiency alternatives found for industry on site (i.e. Natural gas boiler for high temperature heat) in relation to other building categories [118]. In terms of total values, offices uses could predominantly increase the total carbon footprint of the area, what we found to be in line with current trends of high penetration of the Service sector in cities.

#### 4. Conclusions

We presented the City Energy Analyst, an integrated framework for the evaluation of energy efficiency strategies in neighborhoods and city districts in the context of urban transformation. The framework integrates time-dependent methods of supply and demand to a bi-level and multi-objective optimization module for layout of thermal micro-grids and evaluation of multiple conversion alternatives.

The simplified nature of all models used in the framework imply a reduced amount of input variables, low-complexity in return for an acceptable level of uncertainty. In comparison to measured data, the rME and RMSE of most models oscillate between 2% and 5% (see Table 1). The exception lies in the radiation (RMSE = 14%) and the building energy demand models (rME = 32%). Future work will like on improving the reliability of such models. The deterministic models of industrial processes and thermal storage require further validation.

The City Energy Analyst offers the possibility to represent a holistic vision of patterns of demand and supply by means of detailed modeling techniques and a spatiotemporal visualization model. This allows the model to serve as a catalyst for discussions during participatory planning processes and a way of intuitive verification of design approaches.

We presented applications of the model for a neighborhood in Switzerland. We evaluated Pareto optimal energy infrastructure configurations for plausible urban scenarios and conducted a multi-criteria decision analysis to select a tailored configuration.

The results show an integration of 50% to 80% of buildings in thermal micro-grids, integration from 50 to 100% of the available solar potential, a preferable use of photovoltaic electricity, sources of waste heat and centralized use of lake water for cooling purposes. In addition, for a balanced distribution of multi-criteria weights, the results show a relative savings of close to 60% of emissions and 50% of primary energy consumption at a cost of up to 44% higher. For the same balanced distribution, shares of embodied energy in buildings and operation of industrial sectors have the potential to generate the lowest emissions, whereas mobility in the service sector (business flights) potentially generates the highest. In the same way, an increase from today's 16% to 20% efficiency of photovoltaic

technology would allow most scenarios to compile with the local 2000-W/1-t CO<sub>2</sub> per capita target. We conclude the suitability of the framework to assist urban planning authorities looking at options to transform cities for maximum performance.

In the context of scenario comparison, we identified that relative savings up to 15% in emissions and 20% in primary energy with variable costs between up to 23% in the area are more related to the urban design option rather than to its energy systems optimization.

In addition, we compared the performance of the selected configuration to their cross-sectorial (industry, buildings and transportation) and multi-dimensional during construction and operation) environmental impact.

We presented a comprehensive description of focal points of environmental impact as well as a description of the role of industry in a transition towards more sustainable urban environments. In this context, the shares of embodied energy in buildings and operation of industrial sectors have the potential to generate the lowest emissions, whereas mobility in the service sector (business flights) potentially generates the highest. The existence of light industry itself does not jeopardize the future environmental performance of the area, but instead, it brings positive effects as a potential source of waste energy. In contrast top-management levels creates the highest share of emissions due to business flights. Technological advances in energy and building technology will represent a critical factor to attain the 2000-W/1-ton society goals.

In terms of computational time, the framework currently works as a series of decoupled modules. In the future, these modules could be connected to reduce simulation time (i.e. 7 days for 94 buildings during 24 generations starting with 3 individuals and finishing with 1500)<sup>6</sup>. The main bottle-neck is the size and operation optimization process, which increments the number of individuals in every generation. High computational times are followed by the models for optimization of thermal networks, solar resource evaluation, building demand prediction and heat exchangers performance. A major improvement in the computational time could go in the direction of code-parallelization, cloud-computing and reduction of data points (8760 h per building per individual). In this context, Data clustering algorithms and rolling horizon approaches could help to either, respectively, reduce the number of demand points of analysis or improve the process of optimization of the genetic algorithm. The last could have higher advantages as it does not reduce the amount of data analyzed, and thus does not alter the dynamic behavior of the system.

Decision support systems allow for revealing alternatives adaptable to the needs of diverse local stake-holders. Despite of this, a selection of strategies based on maximum budget availability seems a more reasonable selection, especially in contexts with high economic constraints. Both methods could be complemented with relative metrics of costs, which could reveal further opportunities to integrate economically intensive alternatives.

In the Future, the open source framework could be complemented with models for simulation of electric transportation, allowing for the multi-carrier simulation of complete energy systems at the neighborhood scale. This could go in hand with refined models for simulation of outdoor environmental quality, which besides emissions of CO<sub>2</sub>, could address other air and noise pollutants associated to different types of energy production. These models could vary from simple performance factors associated to every energy source up to advanced models of noise propagation. Both models could in principle be developed in Python or another object-oriented language and be constructed as an extension in ArcGIS.

<sup>6</sup> Computer specs: 3.2 GHz quad-core Intel Core i5, 8 GB (two 4 GB) memory. 1 TB hard drive 1. NVIDIA GeForce GT 755M with 1 GB video memory.

In addition, economic variables related to the expected payback time per energy source could reveal opportunities for their further prioritization. Other economic variables such as the marginal value of existing infrastructure is not considered and should be included in the future.

### Acknowledgements

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### Appendix A. Annex A

Figs. 12 and 13

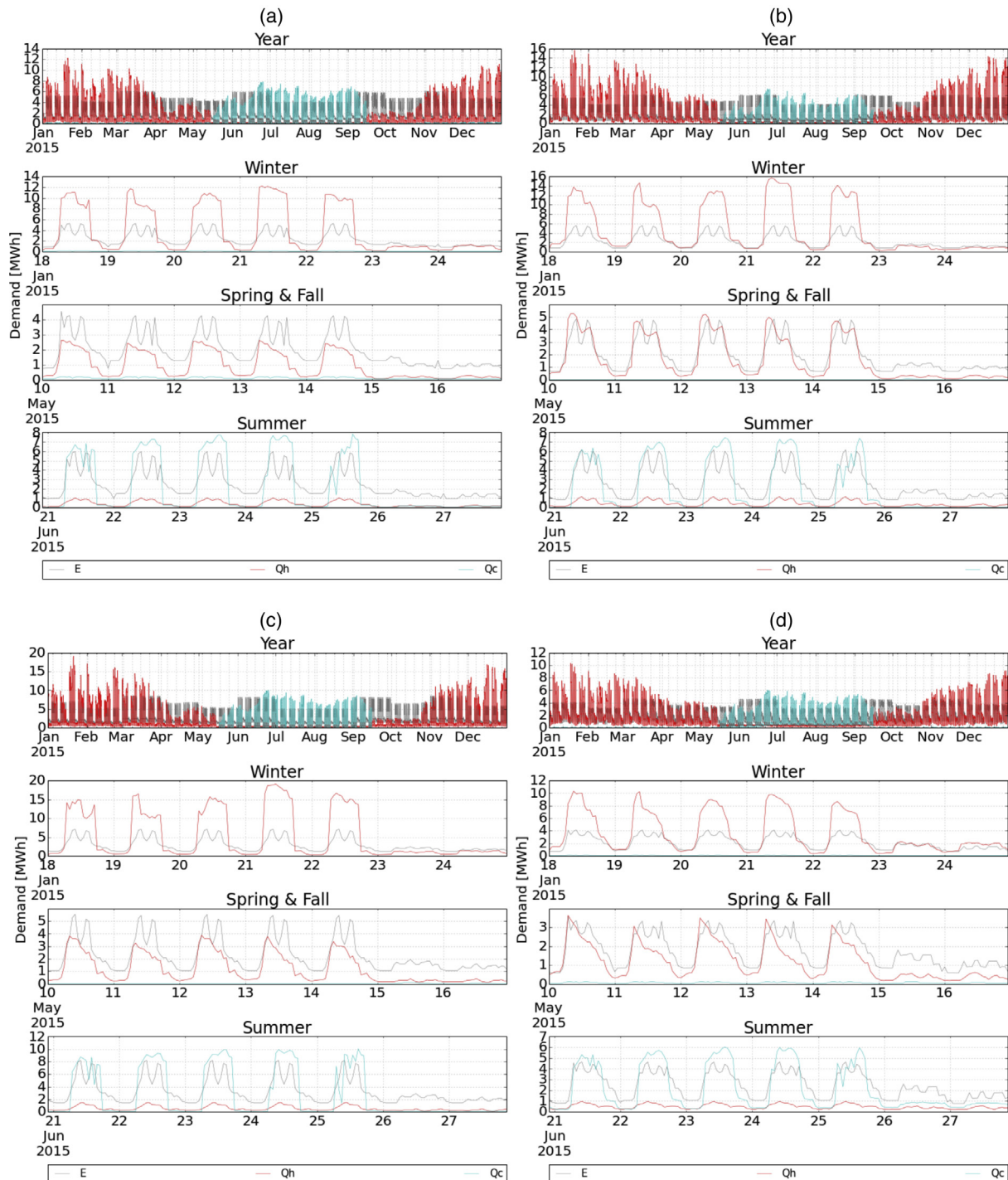


Fig. 12. Energy demand per scenario, (a) BAU, (b) CAMP, (c) HEB, (d) UC.



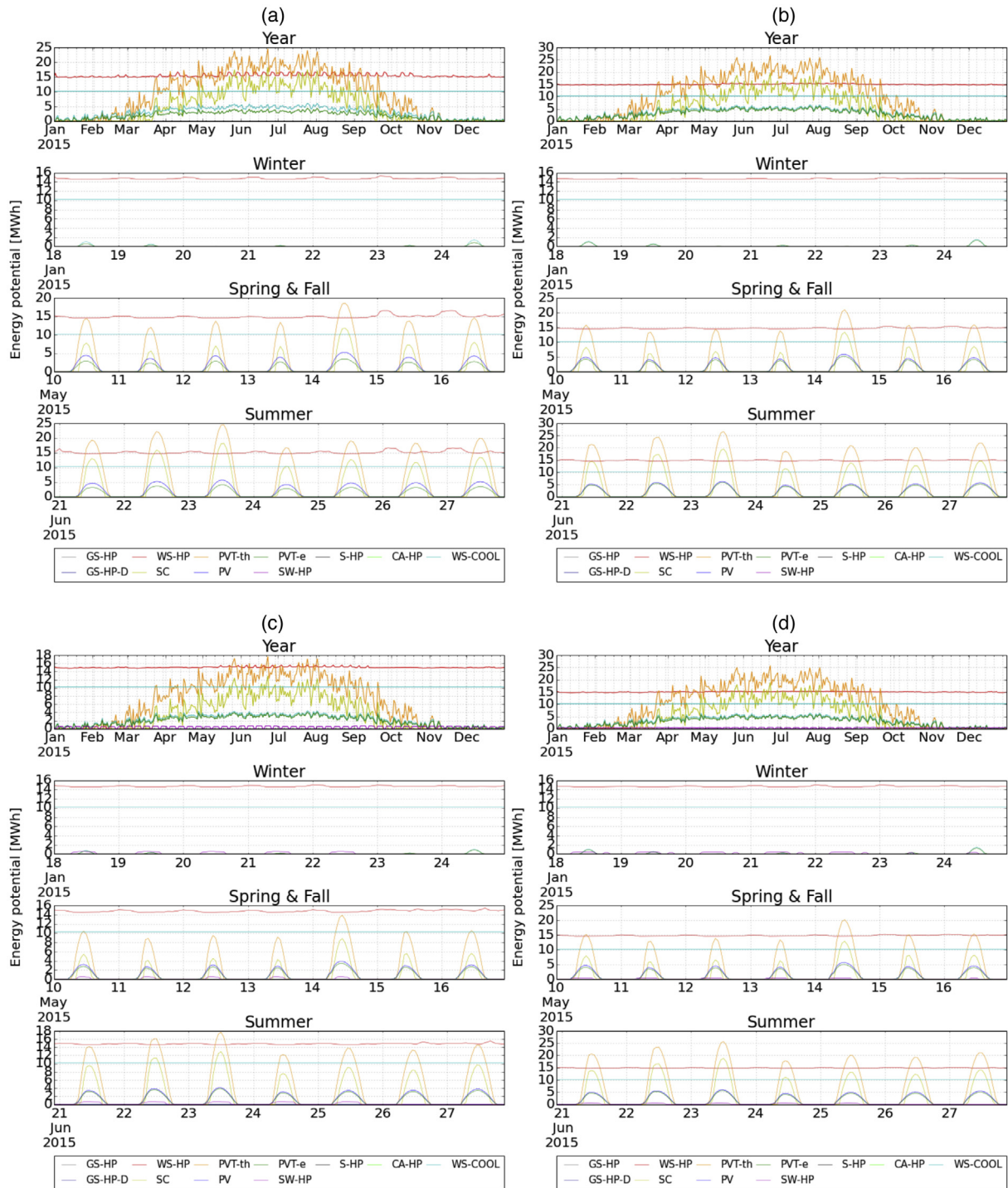


Fig. 13. Energy potential per scenario, (a) BAU, (b) CAMP, (c) HEB, (d) UC.

## Appendix B. Annex B

|                   |                                                                                  |                   |                                                              |
|-------------------|----------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------|
| Variable          |                                                                                  |                   |                                                              |
| $R_{flg}$         | Ground reflectance = 0.2                                                         | $C_{1,data}$      | First order constant servers waste heat = 6.8384E – 08       |
| $(\tau\alpha)_D$  | Average annual incident angle modifier for diffuse radiation = 0.98 rad          | $C_{2,data}$      | Second order constant servers waste heat = 1.8462E – 1       |
| $(\tau\alpha)_C$  | Average annual incident angle modifier for reflected ground radiation = 0.97 rad | $C_{3,data}$      | Third order constant servers waste heat = 1.655E – 3         |
| $\Delta T_{A,HP}$ | Temperature gap at the condenser and evaporator for heat pumps = 5 K             | $C_{4,data}$      | Fourth order constant servers waste heat = 1.3109E – 4       |
| $\rho_w$          | Water density = 998 kg/m <sup>3</sup>                                            | $\Delta T_{HRSG}$ | Temperature pinch at the heat recovery steam generator = 4 K |

|                     |                                                                                            |                         |                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------|
| $\nu_w$             | Water kinematic viscosity (heating/cooling) = (0.404, 1.519) m <sup>2</sup> /s E – 6       | $g$                     | Acceleration due to gravity = 9.81 m/s <sup>2</sup>                                      |
| $\psi_{sc}$         | Linear conductivity of solar collector piping = 0.027 W/m K                                | $\alpha_T$              | Thermal expansion coefficient = 1.6509E – 5 K <sup>-1</sup>                              |
| $L_{mesh}$          | Length of vertices of discretized roof = 2 m                                               | $CD_{surf}$             | Surface drag coefficient = 0.002                                                         |
| $\phi_{sys,lpv}$    | System losses of PV installation = 0.1                                                     | $\rho_a$                | Air density = 1.23 kg/m <sup>3</sup>                                                     |
| $n_{pv}^0$          | Nominal efficiency of PV panel = 0.16                                                      | $\rho_{surf}$           | Density of lake water surface = 998 kg/m <sup>3</sup>                                    |
| $\Delta P_{sc,dis}$ | Linear pressure loss per unit of length of piping for solar instalations = 200 Pa/m        | $Cp_{surf}$             | Specific heat of water surface = 4210 kJ/kg K                                            |
| $\rho_{w/gl}$       | density of the circulating liquid in solar collector (33% glycol) = 1044 kg/m <sup>3</sup> | $\Delta T_{lake}^{max}$ | Maximum change of lake temperature = 0.2 K                                               |
| $Cp_{w/gl}$         | Specific heat capacity mixture of 33% glycol and water = 3680 kJ/kgK                       | $P_{HP}$                | High pressure level in the double pressure steam turbine = 48.2E5 Pa                     |
| $T_{pv}^0$          | Nominal operating temperature of PV panel = 298 K                                          | $P_{LP}$                | Low pressure level in the double pressure steam turbine = 6E5 Pa                         |
| $P_a$               | Atmospheric pressure of air = 101 kPa                                                      | $k_{sw}$                | average coefficient of linear transmittance of the internal sewage network = 0.026 W/m K |
| $P_{ca,sup}$        | Compressed air delivery pressure = 650 kPa                                                 | $Z^0$                   | Vertical depth of the pipe center under the ground = 1.5 m                               |
| $n_{a,pol}$         | Polytropic constant for air = 1.25                                                         | $\rho_g$                | Ground density = 1600 kg/m <sup>3</sup>                                                  |
| $n_{exp}$           | Efficiency of expansion = 0.50                                                             | $Cp_g$                  | heat capacity of the ground = 1300 J/kg K                                                |
| $n_{comp}$          | Efficiency of compressor = 0.95                                                            | $k_g$                   | thermal conductivity of the ground = 1.5 W/m K                                           |
| $C_{1,boiler}$      | First order coefficient condensing boiler = 0.96                                           | $C_{1,sc}$              | First coefficient of transmittance of solar collector = 3.91 W/m <sup>2</sup> K          |
| $C_{2,boiler}$      | Second order coefficient = 1.0                                                             | $C_{2,sc}$              | Second coefficient of transmittance of solar collector = 0.0081 W/m <sup>2</sup> K       |
| $n_{sc}^0$          | Nominal efficiency of solar collector = 0.775                                              | $m_{sc}^0$              | Nominal mass flow rate of solar collector = 0.029 kg/s                                   |
| $\Delta T_{test}$   | Nominal test temperature = 323 K                                                           | $m_{sc}^{min}$          | Minimum mass flow rate of panel = 0.0145 kg/s                                            |
| $T_{w,re}$          | Inlet temperature of the cooling water = 303 K                                             | $m_{sc}^{max}$          | Maximum mass flow rate of panel = 0.0435 kg/s                                            |
| $C_{1,vcc}$         | Rate of internal entropy dissipation = 0.1980 kW/K                                         | $\dot{v}_{sw,a}$        | Average velocity of air in-sewage = 0.15 m/s                                             |
| $C_{2,vcc}$         | Heat leakage = 168.1846 kW                                                                 | $\dot{v}_{sw,w}$        | Average velocity of water in-sewage = 0.9 m/s                                            |
| $C_{3,vcc}$         | Heat exchanger thermal resistance = 0.0201 K/kW                                            | $C_{eff,sc}$            | Thermal inertia coefficient of panel = 8.386 KJ/K                                        |
| $k_{HX}$            | average heat exchanger coefficient = 2500 W/m <sup>2</sup> K                               | $\Delta P_{sc}^0$       | Nominal pressure of solar collector = 170 Pa                                             |
| $k_{p,sw}$          | average coefficient of linear transmittance of the internal sewage network = 0.26 W/m K    | $\Delta P_{sc}^{min}$   | Minimum pressure of solar collector = 80 Pa                                              |
| $k_p$               | average coefficient of linear transmittance of pipe hydraulic network = 0.026 W/m K        | $\Delta P_{sc}^{max}$   | Maximum pressure of solar collector = 270 Pa                                             |
| $h_{HX,sw}$         | Height of heat exchange in sewage = 0.1 m                                                  | $\xi_{el,CH}$           | Price electricity local grid = 0.20 EU/kW h                                              |
| $\delta_m$          | Momentum diffusivity                                                                       | $\xi_{NG,CH}$           | Price natural gas local grid = 0.068 EU/kW h                                             |
| $\delta_T$          | Thermal diffusivity                                                                        | $\xi_{BG,CH}$           | Price Biogas local grid = 0.076 EU/kW h                                                  |
| $n_{FC,th}^0$       | Nominal electrical efficiency of fuel cell (excl. yearly decay) = 0.35                     | $Z_g$                   | Ground depth of heat transfer = 0.40 m                                                   |
| $\nu_w$             | Kinematic viscosity of water = 1.13 m <sup>2</sup> /s E – 3                                | $A_a$                   | Aperture area of solar collector = 2.83 m <sup>2</sup>                                   |
| $b_{sw,p}$          | Wall thickness of sewage pipe = 0.12 m                                                     | $A_p$                   | Aperture area of solar collector = 4.32 m <sup>2</sup>                                   |
| $U_{STO}$           | Heat transfer coefficient of thermal storage = W/m <sup>2</sup> K                          | $n_{FC,e}^0$            | Nominal thermal efficiency of fuel cell (excl. yearly decay) = 0.58                      |

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